INTERVIEW QUESTIONS PYTHON DATA SCIENCE AND MACHINE LEARNING

PYTHON INTERVIEWS QUESTIONS :

1. **What is Python? List some popular applications of Python in the world of technology.**

Ans. Python is a widely-used general-purpose, high-level programming language. It was created by Guido van Rossum in 1991 and further developed by the Python Software Foundation. It was designed with an emphasis on code readability, and its syntax allows programmers to express their concepts in fewer lines of code.
It is used for:

System Scripting
Web Development
Game Development
Software Development
Complex Mathematics

2. **Is Python a compiled language or an interpreted language?**

Ans. Actually, Python is a partially compiled language and partially interpreted language. The compilation part is done first when we execute our code and this will generate byte code internally. This byte code gets converted by the Python virtual machine(p.v.m) according to the underlying platform(machine+operating system).

3. **What is the difference between a Mutable datatype and an Immutable data type?**

Ans. Mutable data types can be edited i.e., they can change at runtime. Eg – List, Dictionary, etc.
Immutable data types can not be edited i.e., they can not change at runtime. Eg – String, Tuple, etc.

4. **What is the difference between / and // in Python?**

Ans. / represents precise division (result is a floating point number) whereas // represents floor division (result is an integer). For Example:

5//2 = 2
5/2 = 2.5

5. **What is List Comprehension? Give an Example.**

Ans. List comprehension is a syntax construction to ease the creation of a list based on an existing iterable.

For Example:

my_list = [i for i in range(1, 10)]

6. **What is a lambda function?**

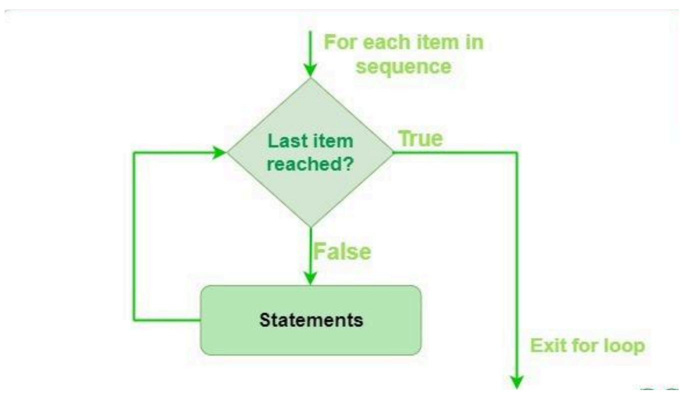
Ans. A lambda function is an anonymous function. This function can have any number of parameters but can have just one statement. For Example:

a = lambda x, y : x*y
print(a(7, 19))

7. **Difference between for loop and while loop in Python**

Ans. In Python, a 'for loop' is used to iterate over a sequence of items, such as a Python tuple, list, string, or range. The loop will execute a block of statements for each item in the sequence.

Python for Loop Flowchart



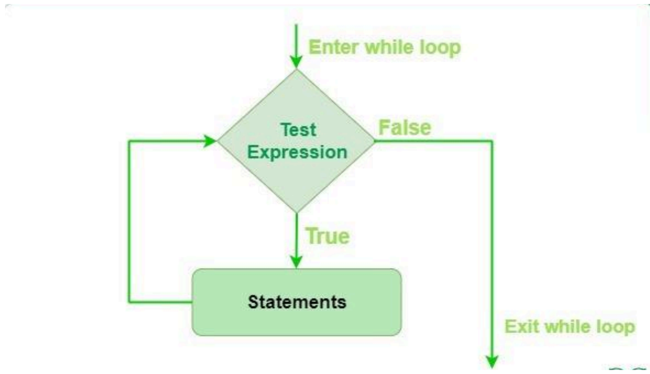
Syntax of Python for loop

for var in iterable:
 # statements

While loop in Python

In Python, a while loop is used to repeatedly execute a block of statements while a condition is true. The loop will continue to run as long as the condition remains true.

Python while Loop Flowchart



Syntax for Python while loop

while condition:

 # Set of statements

8. **What are Built-in data types in Python?**

Ans. The following are the standard or built-in data types in Python:

Numeric: The numeric data type in Python represents the data that has a numeric value. A numeric value can be an integer, a floating number, a Boolean, or even a complex number.
Sequence Type: The sequence Data Type in Python is the ordered collection of similar or different data types.
There are several sequence types in Python:

Python String
Python List
Python Tuple
Python range

Mapping Types: In Python, hashable data can be mapped to random objects using a mapping object. There is currently only one common mapping type, the dictionary, and mapping objects are mutable.
Python Dictionary

Set Types: In Python, a Set is an unordered collection of data types that is iterable, mutable, and has no duplicate elements. The order of elements in a set is undefined though it may consist of various elements.

9. **What are Decorators?**

Ans. Decorators are a very powerful and useful tool in Python as they are the specific change that we make in Python syntax to alter functions easily.

10. What are Iterators in Python?

Ans. In Python, iterators are used to iterate a group of elements, containers like a list. Iterators are collections of items, and they can be a list, tuples, or a dictionary. Python iterator implements __itr__ and the next() method to iterate the stored elements. We generally use loops to iterate over the collections (list, tuple) in Python.

11. What are Generators in Python?

Ans. In Python, the generator is a way that specifies how to implement iterators. It is a normal function except that it yields expression in the function. It does not implement __itr__ and __next__ method and reduces other overheads as well.

If a function contains at least a yield statement, it becomes a generator. The yield keyword pauses the current execution by saving its states and then resumes from the same when required.

12. How is memory management done in Python?

Ans. Python uses its private heap space to manage the memory. Basically, all the objects and data structures are stored in the private heap space. Even the programmer can not access this private space as the interpreter takes care of this space. Python also has an inbuilt garbage collector, which recycles all the unused memory and frees the memory and makes it available to the heap space.

13. What is PIP?

Ans. PIP is an acronym for Python Installer Package which provides a seamless interface to install various Python modules. It is a command-line tool that can search for packages over the internet and install them without any user interaction.

14. What is a zip function?

Ans. Python zip() function returns a zip object, which maps a similar index of multiple containers. It takes an iterable, converts it into an iterator and aggregates the elements based on iterables passed. It returns an iterator of tuples.

15. Explain Python List, Tuple, Dictionary ?

16. What is __init__() in Python?

Ans. The __init__() method is known as a constructor in object-oriented programming (OOP) terminology. It is used to initialize an object's state when it is created. This method is automatically called when a new instance of a class is instantiated.

Purpose:

- Assign values to object properties.
- Perform any initialization operations.

Example:

We have created a `book_shop` class and added the constructor and `book()` function. The constructor will store the book title name and the `book()` function will print the book name.

```
class book_shop:

    # constructor
    def __init__(self, title):
        self.title = title

    # Sample method
    def book(self):
        print('The tile of the book is', self.title)


b = book_shop('Sandman')
b.book()
# The tile of the book is Sandman
```

17. How Would You Generate Random Numbers in Python?

Ans. You must first import the random module to generate random numbers in Python.

The random() function generates a random float value between 0 & 1.

> random.random()

The randrange() function generates a random number within a given range.

Syntax: randrange(beginning, end, step)

Example - > random.randrange(1,10,2)

18. How Will You Check If All the Characters in a String Are Alphanumeric?

Ans. Python has an inbuilt method isalnum() which returns true if all characters in the string are alphanumeric.

Example -

>> "abcd123".isalnum()

Output: True

>>"abcd@123#".isalnum()
Output: False

Another way is to use regex as shown.

>>import re

>>bool(re.match('[A-Za-z0-9]+$','abcd123'))
Output: True

>> bool(re.match('[A-Za-z0-9]+$','abcd@123'))
Output: False

19. Differentiate Between append() and extend().

Ans.

append	extend
append() adds an element to the end of the list .	**extend() adds elements from an iterable to the end of the list**
Example : **>> lst = [1,2,3,]** **>>lst.append(4)** **>>lst** **Output:[1,2,3,4]**	**Example -** **>>lst=[1,2,3]** **>>lst.extend([4,5,6])** **>>lst** **Output:[1,2,3,4,5,6]**

20. Is Python Object-oriented or Functional Programming?

Ans. Python is considered a multi-paradigm language. Python follows the object-oriented paradigm.

Python allows the creation of objects and their manipulation through specific methods.
It supports most of the features of OOPS such as inheritance and polymorphism.
Python follows the functional programming paradigm

Functions may be used as the first-class object.
Python supports Lambda functions which are characteristic of the functional paradigm.

21. What is a Class and an Object ?

Ans. Python Class

A class is considered a blueprint of objects.

We can think of the class as a sketch (prototype) of a house. It contains all the details about the floors, doors, windows, etc.

Based on these descriptions, we build the house; the house is the object.

Since many houses can be made from the same description, we can create many objects from a class.

Define Python Class

We use the class keyword to create a class in Python. For example

```
class ClassName:
    # class definition
```

Here, we have created a class named ClassName

Let’s see an example ,

```
class Bike:
    name = ""
    gear = 0
```

Here,

1. Bike - the name of the class
2. name/gear - variables inside the class with default values "" and 0 respectively.

Note: The variables inside a class are called attributes.

Python Objects

An object is called an instance of a class.

Suppose Bike is a class then we can create objects like bike1, bike2, etc from the class.

Here's the syntax to create an object.

```
objectName = ClassName()
```

Let’s see as example,

```
# create class
class Bike:
    name = ""
    gear = 0

# create objects of class
bike1 = Bike()
```

Here, bike1 is the object of the class. Now, we can use this object to access the class attributes.

22. What is the constructor in a Python class?
23. Write a Python program to check if a string is a palindrome.

```
def is_palindrome(string):
    reversed_string = string[::-1]
    return string == reversed_string

# Test the function
word = "madam"
if is_palindrome(word):
    print(f"{word} is a palindrome")
else:
    print(f"{word} is not a palindrome")
```

24. Write a Python program to find the factorial of a number.

Solution.

```
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n-1)

# Test the function
number = 5
result = factorial(number)
print(f"The factorial of {number} is {result}")
```

25. Write a Python program to find the largest element in a list.

```
def find_largest(numbers):
    largest = numbers[0]
    for num in numbers:
        if num > largest:
            largest = num
    return largest

# Test the function
nums = [10, 5, 8, 20, 3]
largest_num = find_largest(nums)
print(f"The largest number is {largest_num}")
```

26. Write a Python program to check if a number is prime.

```
def is_prime(number):
    if number < 2:
        return False
    for i in range(2, int(number**0.5) + 1):
        if number % i == 0:
```

```python
            return False
    return True

# Test the function
num = 17
if is_prime(num):
    print(f"{num} is a prime number")
else:
    print(f"{num} is not a prime number")
```

27. Write a Python program to find the common elements between two lists.

```python
def find_common_elements(list1, list2):
    common_elements = []
    for item in list1:
        if item in list2:
            common_elements.append(item)
    return common_elements

# Test the function
list_a = [1, 2, 3, 4, 5]
list_b = [4, 5, 6, 7, 8]
common = find_common_elements(list_a, list_b)
print(common)
```

28. Write a Python program to find the second largest number in a list.

```python
def bubble_sort(elements):
    n = len(elements)
    for i in range(n - 1):
        for j in range(n - i - 1):
            if elements[j] > elements[j + 1]:
                elements[j], elements[j + 1] = elements[j + 1], elements[j]

# Test the function
nums = [5, 2, 8, 1, 9]
bubble_sort(nums)
print(nums)
```

29. Write a Python program to remove duplicates from a list.

```python
def remove_duplicates(numbers):
    unique_numbers = []
    for num in numbers:
        if num not in unique_numbers:
            unique_numbers.append(num)
    return unique_numbers
```

```
# Test the function
nums = [1, 2, 3, 2, 1, 3, 2, 4, 5, 4]
unique_nums = remove_duplicates(nums)
print(unique_nums)
```

30. **Write a program to find the Fibonacci series up to a given number.**

```
def remove_duplicates(numbers):
    unique_numbers = []
    for num in numbers:
        if num not in unique_numbers:
            unique_numbers.append(num)
    return unique_numbers

# Test the function
nums = [1, 2, 3, 2, 1, 3, 2, 4, 5, 4]
unique_nums = remove_duplicates(nums)
print(unique_nums)
```

31. **Write a program to convert Celsius to Fahrenheit.**

```
def remove_duplicates(numbers):
    unique_numbers = []
    for num in numbers:
        if num not in unique_numbers:
            unique_numbers.append(num)
    return unique_numbers

# Test the function
nums = [1, 2, 3, 2, 1, 3, 2, 4, 5, 4]
unique_nums = remove_duplicates(nums)
print(unique_nums)
```

32. **Given a positive integer num, write a function that returns True if num is a perfect square else False.**

```
import math

def is_perfect_square(num):
    if num < 0:
        return False
    sqrt = math.isqrt(num)
    return sqrt * sqrt == num

# Test cases
```

```
print(is_perfect_square(16))  # True
print(is_perfect_square(14))  # False
```

DATASCIENCE INTERVIEWS QUESTIONS :

1. **What are the advantages of NumPy over regular Python lists?**

Memory

Numpy arrays consume less memory.

For example, if you create a list and a Numpy array of a thousand elements. The list will consume 48K bytes, and the Numpy array will consume 8k bytes of memory.

Speed

Numpy arrays take less time to perform the operations on arrays than lists.

For example, if we are multiplying two lists and two Numpy arrays of 1 million elements together. It took 0.15 seconds for the list and 0.0059 seconds for the array to operate.

Vesititly

Numpy arrays are convenient to use as they offer simple array multiple, addition, and a lot more built-in functionality. Whereas Python lists are incapable of running basic operations.

2. **What is the difference between merge, join and concatenate?**

Merge

Merge two DataFrames named series objects using the unique column identifier.

It requires two DataFrame, a common column in both DataFrame, and “how” you want to join them together. You can left, right, outer, inner, and cross join two data DataFrames. By default, it is an inner join.

```
pd.merge(df1, df2, how='outer', on='Id')
```

Join

Join the DataFrames using the unique index. It requires an optional `on` argument that can be a column or multiple column names. By default, the join function performs a left join.

```
df1.join(df2)
```

Concatenate

Concatenate joins two or multiple DataFrames along a particular axis (rows or columns). It doesn't require an `on` argument.

```
pd.concat(df1,df2)
```

- **join():** it combines two DataFrames by index.
- **merge():** it combines two DataFrames by the column or columns you specify.
- **concat():** it combines two or more DataFrames vertically or horizontally.

3. **How do you identify and deal with missing values?**

Identifying missing values

We can identify missing values in the DataFrame by using the `isnull()` function and then applying `sum()`. `Isnull()` will return boolean values, and the sum will give you the number of missing values in each column.

In the example, we have created a dictionary of lists and converted it into a pandas DataFrame. After that, we used isnull().sum() to get the number of missing values in each column.

```
import pandas as pd
import numpy as np

# dictionary of lists
dict = {'id':[1, 4, np.nan, 9],
        'Age': [30, 45, np.nan, np.nan],
        'Score':[np.nan, 140, 180, 198]}

# creating a DataFrame
df = pd.DataFrame(dict)

df.isnull().sum()
# id       1
# Age      2
# Score    1
```

Dealing with missing values

1. There are various ways of dealing with missing values.

2. Drop the entire row or the columns if it consists of missing values using `dropna()`. This method is not recommended, as you will lose important information.
3. Fill the missing values with the constant, average, backward fill, and forward fill using the `fillna()` function.
4. Replace missing values with a constant String, Integer, or Float using the `replace()` function.
5. Fill in the missing values using an interpolation method.

Note: make sure you are working with a larger dataset while using the `dropna()` function.

```
# drop missing values
```

```python
df.dropna(axis = 0, how ='any')

#fillna
df.fillna(method ='bfill')

#replace null values with -999
df.replace(to_replace = np.nan, value = -999)

# Interpolate
df.interpolate(method ='linear', limit_direction ='forward')
```

4. **Which all Python libraries have you used for visualization?**

Data visualization is the most important part of data analysis. You get to see your data in action, and it helps you find hidden patterns.

The most popular Python data visualization libraries are:

1. Matplotlib
2. Seaborn
3. Plotly

In Python, we generally use Matplotlib and seaborn to display all types of data visualization. With a few lines of code, you can use it to display scatter plot, line plot, box plot, bar chart, and many more.

For interactive and more complex applications, we use Plotly. You can use it to create colorful interactive graphs with a few lines of code. You can zoom, apply animation, and even add control functions. Plotly provides more than 40 unique types of charts, and we can even use them to create a web application or dashboard.

5. **The difference between Data Mining and Data Profiling ?**

Data Mining	Data Profiling
Data mining is the process of discovering relevant information that has not been identified before .	**Data profiling is done to evaluate a dataset for its uniqueness , logic , and consistency.**
In data mining , raw data is converted into valuable information.	**It cannot identify inaccurate or incorrect data values.**

6. **Defining the term Data Wrangling ?**

Data wrangling is the process of transforming and mapping data from one data form into another format.

7. **What are the various steps involved in any analytics project ?**

8. What are the common problems data analysts encounter during analysis ?

- Handling Redundant Data
- Collecting the Right Data at the Right Time
- HandlingData Purging and Storage Problems
- Keeping Data Secure
- Handling Compliance

9. What is the significance of Exploratory Data Analysis (EDA) ?

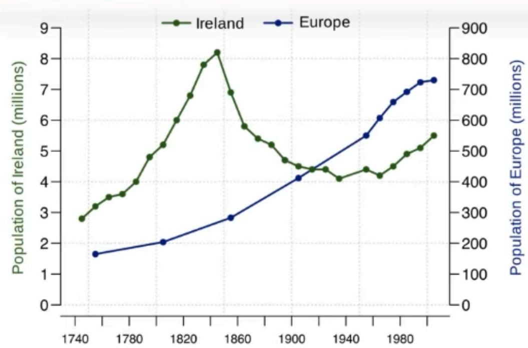
- Better understanding of data
- Have more confidence in your decisions
- Refine feature selection during modeling
- Discover hidden data trends

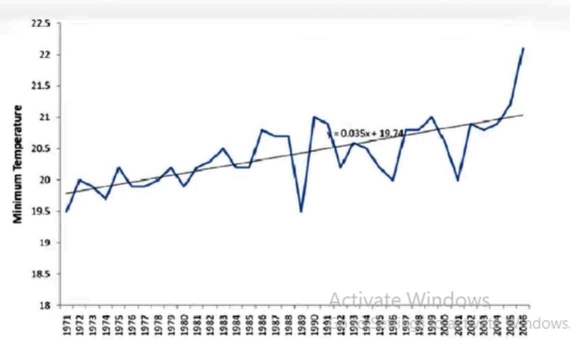
10. What are the best methods for cleaning ?

- Create a data cleaning plan by understanding where the common errors take place and keep all the communications open
- Before working with the data, identify and remove the duplicates. This will lead to an easy and effective data analysis process.
- Focus on the accuracy of the data. Set cross-field validation , maintain the value type of data and provide mandatory constraints.
- Normalize the data at the entry point so that it is less chaotic. You will be able to ensure that all information is standardised, leading to fewer errors on entry.

11. How to handle missing values in a dataset ?
12. What is Time Series Analysis ?

Time Series analysis is a statistical procedure that deals with the ordered sequence of values of a variable at equally spaced time interval.





13. How do you treat Outliers in a Dataset ?

An outlier is a data point that is distant from other similar points. They may be due to variability in the measurement or may indicate experimental errors.

- Drop Outliers
- Assign a new value
- Cap Outliers Data
- Try a new Transfomation

14. **What is the correct syntax for reshape() function in Numpy ?**

reshape(array, shape)

15. **What are the different ways to create a data frame in Pandas ?**

There are 2 ways to Create a DataFrame in Pandas.

- **Initialising a List**
- **Initialising a Dictionary**

16. **There are two arrays , ‘A’ and ‘B’ . Stack the arrays A and B horizontally using the Numpy library in Python.**

There are 2 ways to do this :

- **Concatenate Method → np.concatenate([a,b], axis=1)**
- **Hstack Method → np.hstack([a,b])**

17. **How can you add a column to a Pandas Dataframe ?**

Df.[‘column_name’] = list

18. **How will you print four random integers between 1 and 15 using Numpy ?**

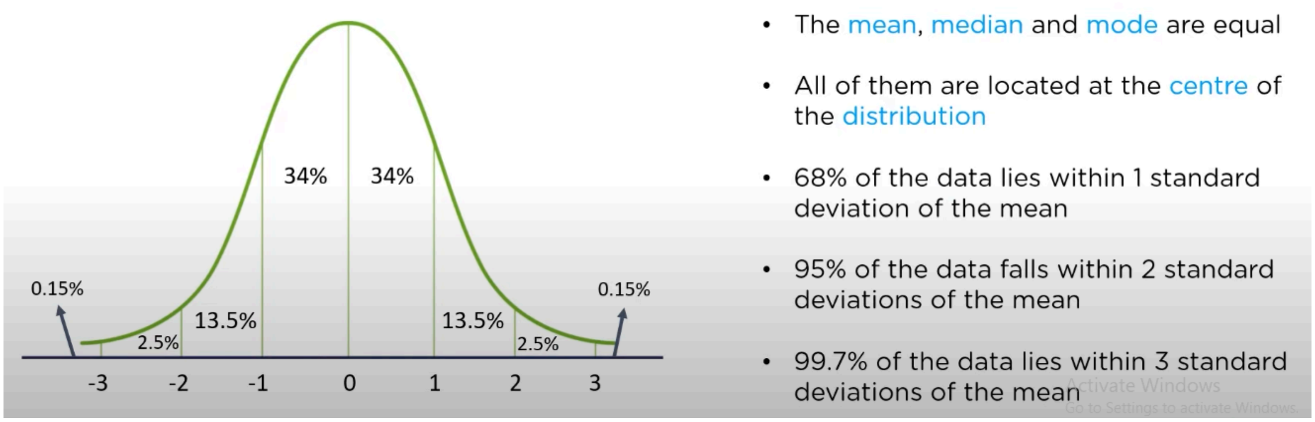
Import numpy as np
rand_arr = np.random.randint(1,15,4)
print(“\n Random number from 1 to 15 are”, rand_arr)

19. **How to select specific columns from a Dataframe ?**

Df[[‘column_name_1’, ‘column_name_2,----]]

20. **What do you understand by the term of normal distribution ?**

Normal Distribution is a type of continuous probability that is symmetric about the mean and in a graph, normal distribution will appear as a bell curve .



21. **How is overfitting different from underfitting ?**

Overfitting	**Underfitting**
Model trains well using the training dataset.	**Model does not train well on the training dataset.**
Performance takes a deep dive on testing dataset.	**Performs poorly on both training and testing dataset.**

Happens when your model learns random fluctuations and noise in training dataset.	**The training dataset is small. Developing linear model with non linear dataset.**

22. How will you select the Department and Age columns from an Employee dataframe ?

```python
import pandas as pd

# Intialise data of lists.
data = {'Name':['Nick', 'Ricky', 'Mathew', 'Andrew'], 'Age':[30, 42, 45, 35],
        'Department':['Manufacturing', 'IT', 'Marketing', 'Sales']}

# Create the DataFrame
Employees = pd.DataFrame(data)
```

```python
# Print the output.
Employees
```

	Name	Age	Department
0	Nick	30	Manufacturing
1	Ricky	42	IT
2	Mathew	45	Marketing
3	Andrew	35	Sales

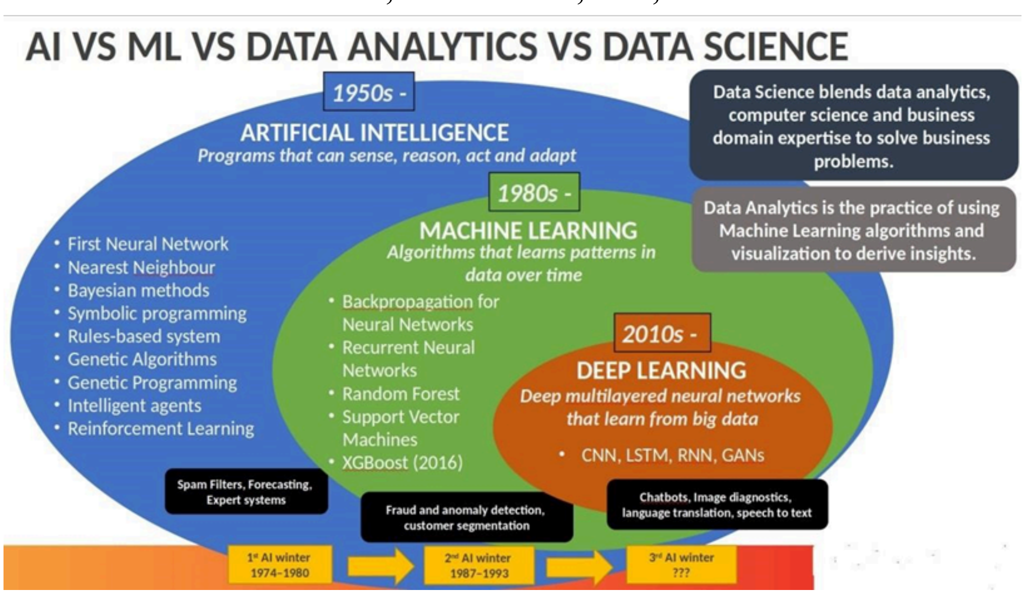
To select Department and Age from the dataframe

```python
Employees[['Department', 'Age']]
```

	Department	Age
0	Manufacturing	30
1	IT	42
2	Marketing	45
3	Sales	35

MACHINE LEARNING INTERVIEWS QUESTIONS :

1. What is the difference between AI, Data Science, ML, and DA ?



Artificial Intelligence: AI is purely math and scientific exercise, but when it became computational, it started to solve human problems formalized into a subset of computer science. Artificial intelligence has changed the original computational statistics paradigm to the modern idea that machines could mimic actual human capabilities, such as decision making and performing more "human" tasks. Modern AI into two categories

1. General AI - Planning, decision making, identifying objects, recognizing sounds, social & business transactions
2. Applied AI - driverless/ Autonomous car or machine smartly trade stocks

Machine Learning: Instead of engineers "teaching" or programming computers to have what they need to carry out tasks, that perhaps computers could teach themselves – learn something without being explicitly programmed to do so. ML is a form of AI where based on more data, and they can change actions and response, which will make more efficient, adaptable and scalable. e.g., navigation apps and recommendation engines. Classified into:-

1. Supervised
2. Unsupervised
3. Reinforcement learning

Data Science: Data science has many tools, techniques, and algorithms called from these fields, plus others –to handle big data

The goal of data science, somewhat similar to machine learning, is to make accurate predictions and to automate and perform transactions in real-time, such as purchasing internet traffic or automatically generating content.

Data science relies less on math and coding and more on data and building new systems to process the data. Relying on the fields of data integration, distributed architecture, automated machine learning, data visualization, data engineering, and automated data-driven decisions, data science can cover an entire spectrum of data processing, not only the algorithms or statistics related to data.

Deep Learning: It is a technique for implementing ML.

ML provides the desired output from a given input, but DL reads the input and applies it to other data. In ML, we can easily classify the flower based upon the features. Suppose you want a machine to look at an image and determine what it represents to the human eye, whether a face, flower, landscape, truck, building, etc.

Machine learning is not sufficient for this task because machine learning can only produce an output from a data set – whether according to a known algorithm or based on the inherent structure of the data. You might be able to use machine learning to determine whether an image was of an "X" – a flower, say – and it would learn and get more accurate. But that output is binary (yes/no) and is dependent on the algorithm, not the data. In the image recognition case, the outcome is not binary and not dependent on the algorithm. The neural network performs MICRO calculations with computations on many layers. Neural networks also support weighting data for 'confidence. This results in a probabilistic system, vs. deterministic, and can handle tasks that we think of as requiring more 'human-like' judgement.

2. **What is the difference between Supervised learning , Unsupervised learning and Reinforcement learning ?**

Machine Learning

Machine learning is the scientific study of algorithms and statistical models that computer systems use to effectively perform a specific task without using explicit instructions, relying on patterns and inference instead.

Building a model by learning the patterns of historical data with some relationship between data to make a data-driven prediction.

Types of Machine Learning

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Supervised learning

In a supervised learning model, the algorithm learns on a labeled dataset, to generate reasonable predictions for the response to new data. (Forecasting outcome of new data)

- Regression
- Classification

Unsupervised learning

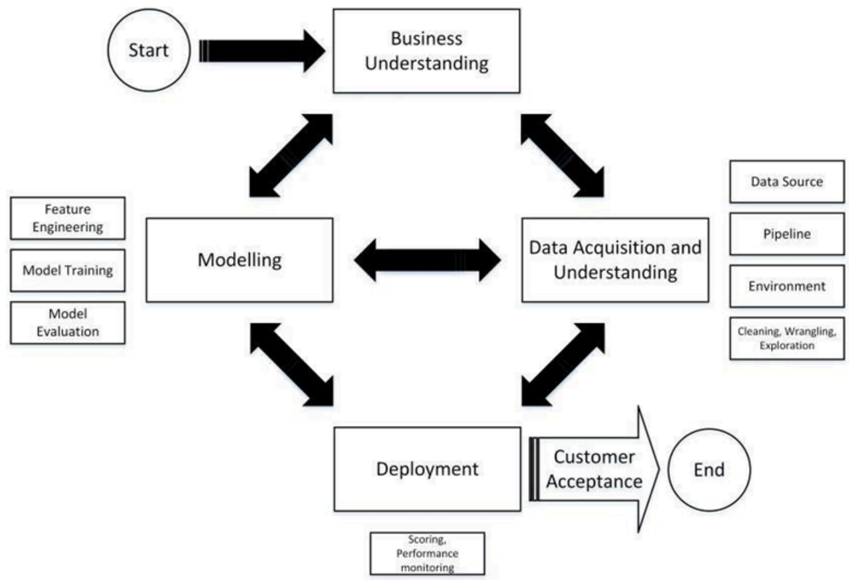
An unsupervised model, in contrast, provides unlabelled data that the algorithm tries to make sense of by extracting features, co-occurrence and underlying patterns on its own. We use unsupervised learning for

- Clustering
- Anomaly detection
- Association
- Autoencoders

Reinforcement Learning

Reinforcement learning is less supervised and depends on the learning agent in determining the output solutions by arriving at different possible ways to achieve the best possible solution.

3. **Describe the general architecture of Machine Learning ?**



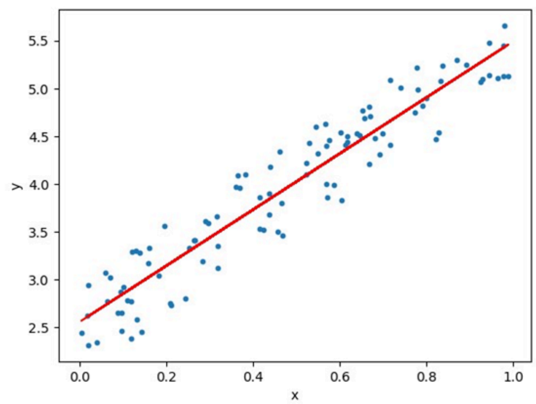
Business understanding: Understand the give use case, and also, it's good to know more about the domain for which the use cases are built.

Data Acquisition and Understanding: Data gathering from different sources and understanding the data. Cleaning the data, handling the missing data if any, data wrangling, and EDA(Exploratory data analysis).

Modeling: Feature Engineering - scaling the data, feature selection - not all features are important. We use the backward elimination method, correlation factors, PCA and domain knowledge to select the features.
Model Training based on trial and error method or by experience, we select the algorithm and train with the selected features.
Model evaluation Accuracy of the model , confusion matrix and cross-validation.
If accuracy is not high, to achieve higher accuracy, we tune the model...either by changing the algorithm used or by feature selection or by gathering more data, etc.
Deployment - Once the model has good accuracy, we deploy the model either in the cloud or Rasberry py or any other place. Once we deploy, we monitor the performance of the model.if its good...we go live with the model or reiterate the all process until our model performance is good.
It's not done yet!!!
What if, after a few days, our model performs badly because of new data. In that case, we do all the process again by collecting new data and redeploy the model.

4. **What is Linear Regression ?**

Linear Regression tends to establish a relationship between a dependent variable(Y) and one or more independent variable(X) by finding the best fit of the straight line.
The equation for the Linear model is Y = mX+c, where m is the slope and c is the intercept



In the above diagram, the blue dots we see are the distribution of 'y' w.r.t 'x.' There is no straight line that runs through all the data points. So, the objective here is to fit the best fit of a straight line that will try to minimize the error between the expected and actual value.

5. **What is R Square ? Where to use and where not ?**

R-squared is a statistical measure of how close the data are to the fitted regression line. It is also known as the coefficient of determination, or the coefficient of multiple determination for multiple regression.

The definition of R-squared is the percentage of the response variable variation that is explained by a linear model.

R-squared = Explained variation / Total variation

R-squared is always between 0 and 100%.

0% indicates that the model explains none of the variability of the response data around its mean.

100% indicates that the model explains all the variability of the response data around its mean.

In general, the higher the R-squared, the better the model fits your data.

Sum Squared Regression Error

$$R^2 = 1 - \frac{SS_{Regression}}{SS_{Total}}$$

Sum Squared Total Error

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

There is a problem with the R-Square. The problem arises when we ask this question to ourselves.** Is it good to help as many independent variables as possible?**

The answer is No because we understood that each independent variable should have a meaningful impact. But, even** if we add independent variables which are not meaningful**, will it improve R-Square value?

Yes, this is the basic problem with R-Square. How many junk independent variables or important independent variable or impactful independent variable you add to your model, the R-Squared value will always increase. It will never decrease with the addition of a newly independent variable, whether it could be an impactful, non-impactful, or bad variable, so we need another way to measure equivalent RSquare, which penalizes our model with any junk independent variable.

So, we calculate the **Adjusted R-Square** with a better adjustment in the formula of generic R-square.

$$R^2_{adjusted} = 1 - \frac{(1 - R^2)(N - 1)}{N - p - 1}$$

where

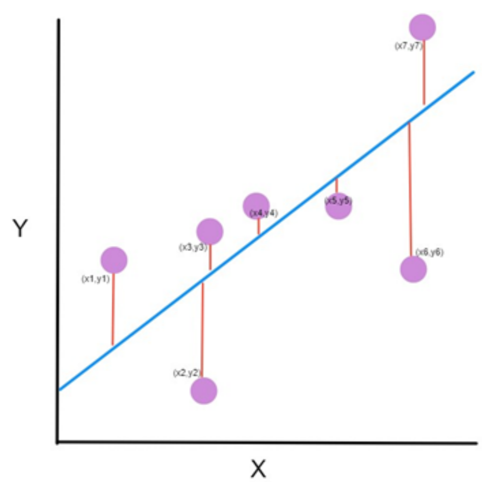
R^2 = sample R-square

p = Number of predictors

N = Total sample size.

6. **What is Mean Square Error ?**

The mean squared error tells you how close a regression line is to a set of points. It does this by taking the distances from the points to the regression line (these distances are the “errors”) and squaring them.



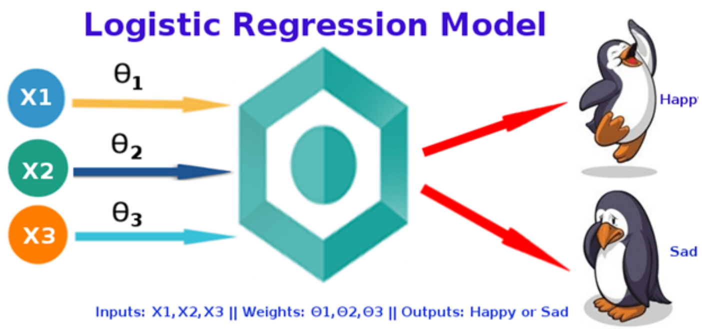
The line equation is y=Mx+B. We want to find M (slope) and B (y-intercept) that minimizes the squared error.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{y}_i)^2$$

7. What is Logistic Regression?

Answer:

The logistic regression technique involves the dependent variable, which can be represented in the binary (0 or 1, true or false, yes or no) values, which means that the outcome could only be in either one form of two. For example, it can be utilized when we need to find the probability of a successful or fail event.



Logistic Regression is used when the dependent variable (target) is categorical.

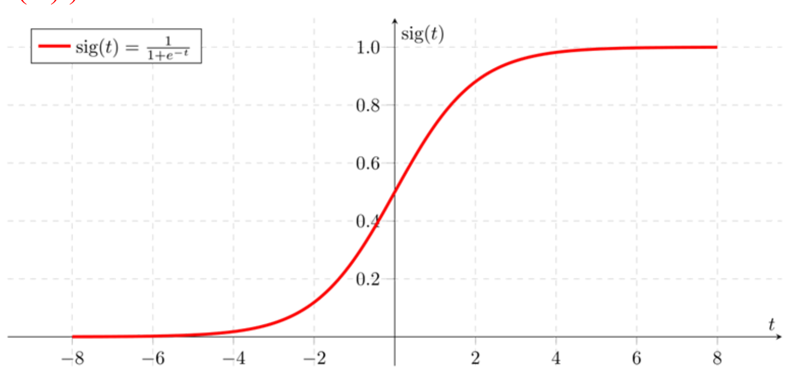
Model

Output = 0 or 1

Z = WX + B

hΘ(x) = sigmoid (Z)

hΘ(x) = log(P(X) / 1 - P(X)) = WX +B



If ‘Z’ goes to infinity, Y(predicted) will become 1, and if ‘Z’ goes to negative infinity, Y(predicted) will become 0.

The output from the hypothesis is the estimated probability. This is used to infer how confident can predicted value be actual value when given an input X.

Cost Function

Cost (hΘ(x) , Y(Actual)) = -log (hΘ(x)) if y=1

-log (1 - hΘ(x)) if y=0

This implementation is for binary logistic regression. For data with more than 2 classes, softmax re gression has to be used.

8. Difference between logistic and linear regression?

Answer:

Linear and Logistic regression are the most basic form of regression which are commonly used. The essential difference between these two is that Logistic regression is used when the dependent variable is binary. In contrast, Linear regression is used when the dependent variable is continuous, and the nature of the regression line is linear.

Key Differences between Linear and Logistic Regression

Linear regression models data using continuous numeric value. As against, logistic regression models the data in the binary values.

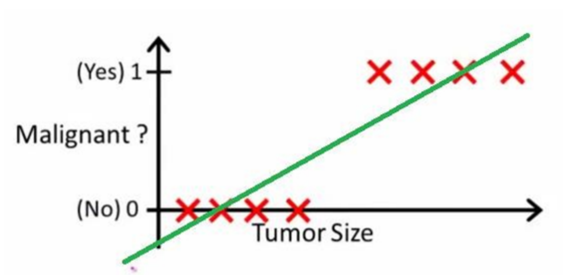
Linear regression requires to establish the linear relationship among dependent and independent variables, whereas it is not necessary for logistic regression.

In linear regression, the independent variable can be correlated with each other. On the contrary, in the logistic regression, the variable must not be correlated with each other.

9. Why can't we do a classification problem using Regression?

Answer:-

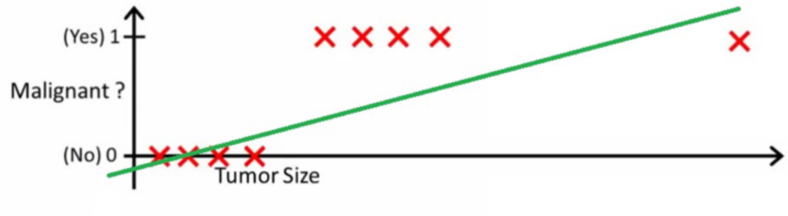
With linear regression you fit a polynomial through the data - say, like on the example below, we fit a straight line through {tumor size, tumor type} sample set:



Above, malignant tumors get 1, and non-malignant ones get 0, and the green line is our hypothesis h(x). To make predictions, we may say that for any given tumor size x, if h(x) gets bigger than 0.5, we predict malignant tumors. Otherwise, we predict benignly.

It looks like this way, we could correctly predict every single training set sample, but now let's change the task a bit.

Intuitively it's clear that all tumors larger certain threshold are malignant. So let's add another sample with huge tumor size, and run linear regression again:



Now our h(x)>0.5→malignant doesn't work anymore. To keep making correct predictions, we need to change it to h(x)>0.2 or something - but that not how the algorithm should work.

We cannot change the hypothesis each time a new sample arrives. Instead, we should learn it off the training set data, and then (using the hypothesis we've learned) make correct predictions for the data we haven't seen before. Linear regression is unbounded.

10. Entropy, Information Gain, Gini Index, Reducing Impurity?

Answer:

There are different attributes which define the split of nodes in a decision tree. There are few algorithms to find the optimal split.

1) ID3(Iterative Dichotomiser 3): This solution uses Entropy and Information gain as metrics to form a better decision tree. The attribute with the highest information gain is used as a root node, and a similar approach is followed after that. Entropy is the measure that characterizes the impurity of an arbitrary collection of examples.

Entropy

Entropy $H(S)$ is a measure of the amount of uncertainty in the (data) set S (i.e. entropy characterizes the (data) set S).

$$H(S) = \sum_{c \in C} -p(c) \log_2 p(c)$$

Where,

- S – The current (data) set for which entropy is being calculated (changes every iteration of the ID3 algorithm)
- C – Set of classes in S C={ yes, no }
- $p(c)$ – The proportion of the number of elements in class c to the number of elements in set S

When $H(S) = 0$, the set S is perfectly classified (i.e. all elements in S are of the same class).

In ID3, entropy is calculated for each remaining attribute. The attribute with the **smallest** entropy is used to split the set S on this iteration. The higher the entropy, the higher the potential to improve the classification here.

Entropy varies from 0 to 1. 0 if all the data belong to a single class and 1 if the class distribution is equal. In this way, entropy will give a measure of impurity in the dataset.

Steps to decide which attribute to split:

1. Compute the entropy for the dataset
2. For every attribute:

2.1 Calculate entropy for all categorical values.

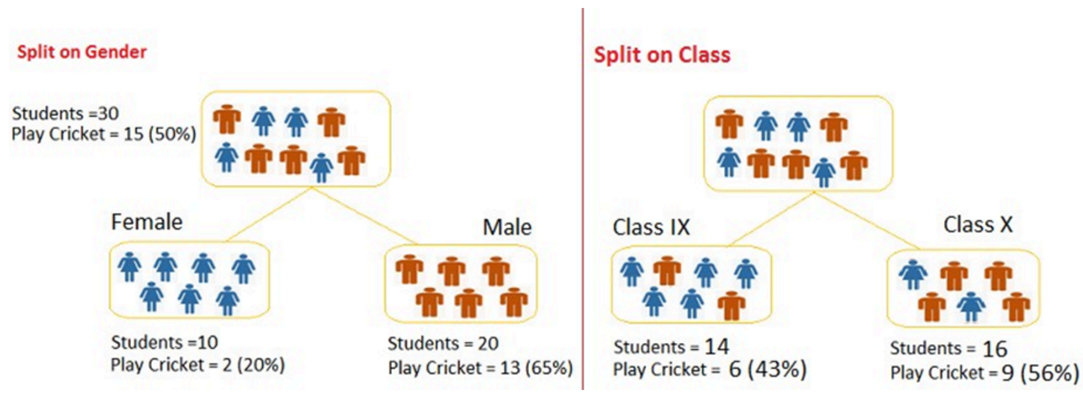
2.2 Take average information entropy for the attribute.

2.3 Calculate gain for the current attribute.
3. Pick the attribute with the highest information gain.
4. Repeat until we get the desired tree.

A leaf node is decided when entropy is zero
Information Gain = 1 - ∑ (Sb/S)*Entropy (Sb)
Sb - Subset, S - entire data

2) CART Algorithm (Classification and Regression trees): In CART, we use the GINI index as a metric. Gini index is used as a cost function to evaluate split in a dataset Steps to calculate Gini for a split:
1. Calculate Gini for subnodes, using formula sum of the square of probability for success and failure (p2+q2).
2. Calculate Gini for split using weighted Gini score of each node of that split.

Choose the split based on higher Gini value



Split on Gender:
Gini for sub-node Female = (0.2)*(0.2)+(0.8)*(0.8)=0.68
Gini for sub-node Male = (0.65)*(0.65)+(0.35)*(0.35)=0.55
Weighted Gini for Split Gender = (10/30)*0.68+(20/30)*0.55 = 0.59

Similar for Split on Class:
Gini for sub-node Class IX = (0.43)*(0.43)+(0.57)*(0.57)=0.51 Gini for sub-node Class X = (0.56)*(0.56)+(0.44)*(0.44)=0.51
Weighted Gini for Split Class = (14/30)*0.51+(16/30)*0.51 = 0.51

Here Weighted Gini is high for gender, so we consider splitting based on gender

11. How to control leaf height and Pruning?

Answer:

To control the leaf size, we can set the parameters:-

1. Maximum depth :

Maximum tree depth is a limit to stop the further splitting of nodes when the specified tree depth has been reached during the building of the initial decision tree.

NEVER use maximum depth to limit the further splitting of nodes. In other words: use the largest possible value.

2. Minimum split size:

Minimum split size is a limit to stop the further splitting of nodes when the number of observations in the node is lower than the minimum split size.

This is a good way to limit the growth of the tree. When a leaf contains too few observations, further splitting will result in overfitting (modeling of noise in the data).

3. Minimum leaf size

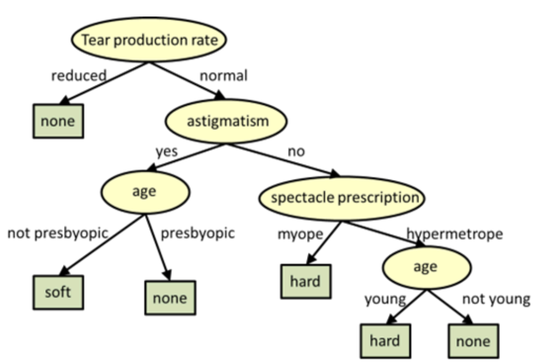
Minimum leaf size is a limit to split a node when the number of observations in one of the child nodes is lower than the minimum leaf size.

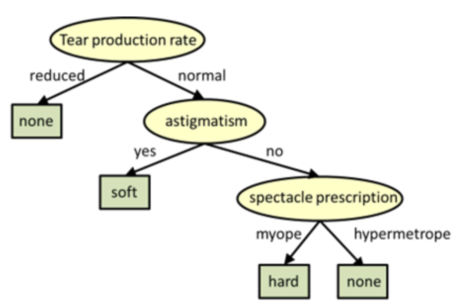
Pruning is mostly done to reduce the chances of overfitting the tree to the training data and reduce the overall complexity of the tree.

There are two types of pruning: Pre-pruning and Post-pruning.

1. **Pre-pruning** is also known as the early stopping criteria. As the name suggests, the criteria are set as parameter values while building the model. The tree stops growing when it meets any of these pre-pruning criteria, or it discovers the pure classes.
2. **In Post-pruning**, the idea is to allow the decision tree to grow fully and observe the CP value. Next, we prune/cut the tree with the optimal CP(Complexity Parameter) value as the parameter.

The CP (complexity parameter) is used to control tree growth. If the cost of adding a variable is higher, then the value of CP, tree growth stops.





12. **What is the Variance and Bias tradeof ?**

Answer:

In predicting models, the prediction error is composed of two different errors

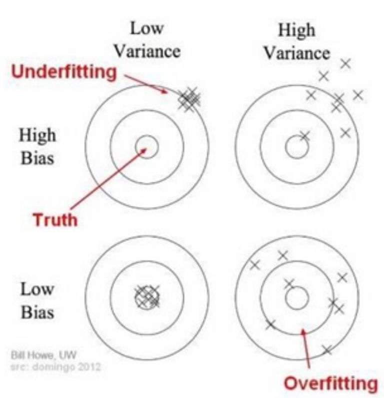
1. Bias

2. Variance

It is important to understand the variance and bias trade-off which tells about to minimize the Bias and Variance in the prediction and avoids overfitting & under fitting of the model.

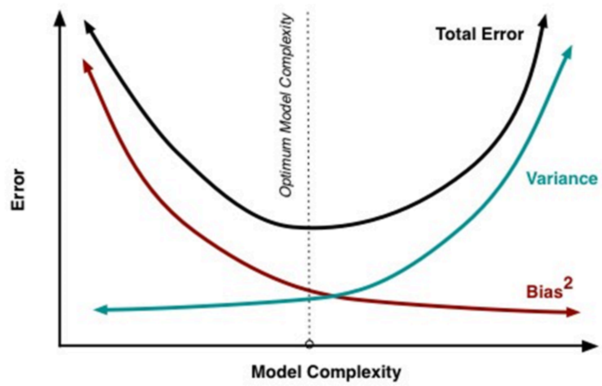
Bias: It is the difference between the expected or average prediction of the model and the correct value which we are trying to predict. Imagine if we are trying to build more than one model by collecting different data sets, and later on, evaluating the prediction, we may end up by different prediction for all the models. So, bias is something which measures how far these model prediction from the correct prediction. It always leads to a high error in training and test data.

Variance: Variability of a model prediction for a given data point. We can build the model multiple times, so the variance is how much the predictions for a given point vary between different realizations of the model.



For example: Voting Republican - 13 Voting Democratic - 16 Non-Respondent - 21 Total - 50
The probability of voting Republican is 13/(13+16), or 44.8%. We put out our press release that the Democrats are going to win by over 10 points; but, when the election comes around, it turns out they lose by 10 points. That certainly reflects poorly on us. Where did we go wrong in our model?
Bias scenario's: using a phonebook to select participants in our survey is one of our sources of bias. By only surveying certain classes of people, it skews the results in a way that will be consistent if we repeated the entire model building exercise. Similarly, not following up with respondents is another source of bias, as it consistently changes the mixture of responses we get. On our bulls-eye diagram, these move us away from the center of the target, but they would not result in an increased scatter of estimates.

Variance scenarios: the small sample size is a source of variance. If we increased our sample size, the results would be more consistent each time we repeated the survey and prediction. The results still might be highly inaccurate due to our large sources of bias, but the variance of predictions will be reduced



13. **What are Ensemble Methods ?**

Answer

1. Bagging and Boosting

Decision trees have been around for a long time and also known to suffer from bias and variance. You will have a large bias with simple trees and a large variance with complex trees.

Ensemble methods - which combines several decision trees to produce better predictive performance than utilizing a single decision tree. The main principle behind the ensemble model is that a group of weak learners come together to form a strong learner.

Two techniques to perform ensemble decision trees:

1. **Bagging**
2. **Boosting**

Bagging (Bootstrap Aggregation) is used when our goal is to reduce the variance of a decision tree. Here the idea is to create several subsets of data from the training sample chosen randomly with replacement. Now, each collection of subset data is used to train their decision trees. As a result, we end up with an ensemble of different models. Average of all the predictions from different trees are used which is more robust than a single decision tree.

Boosting is another ensemble technique to create a collection of predictors. In this technique, learners are learned sequentially with early learners fitting simple models to the data and then analyzing data for errors. In other words, we fit consecutive trees (random sample), and at every step, the goal is to solve for net error from the prior tree.
When a hypothesis misclassifies an input, its weight is increased, so that the next hypothesis is more likely to classify it correctly. By combining the whole set at the end converts weak learners into a better performing model.

The different types of boosting algorithms are:

1. **AdaBoost**
2. **Gradient Boosting**
3. **XGBoost**

14. **What is the Confusion Matrix?**

Answer:

A confusion matrix is a table that is often used to describe the performance of a classification model (or "classifier") on a set of test data for which the true values are known. It allows the visualization of the performance of an algorithm.

A confusion matrix is a summary of prediction results on a classification problem. The number of correct and incorrect predictions are summarized with count values and broken down by each class.

This is the key to the confusion matrix.
It gives us insight not only into the errors being made by a classifier but, more importantly, the types of errors that are being made.

	Class 1 Predicted	Class 2 Predicted
Class 1 Actual	TP	FN
Class 2 Actual	FP	TN

Here,

- Class 1: Positive
- Class 2: Negative Definition of the Terms:

1. **Positive (P):** Observation is positive (for example: is an apple).
2. **Negative (N):** Observation is not positive (for example: is not an apple).
3. **True Positive (TP)**: Observation is positive, and is predicted to be positive.
4. **False Negative (FN):** Observation is positive, but is predicted negative.
5. **True Negative (TN):** Observation is negative, and is predicted to be negative.
6. **False Positive (FP):** Observation is negative, but is predicted positive.

15. **What are F1 Score, precision and recall?**

Recall:-

Recall can be defined as the ratio of the total number of correctly classified positive examples divide to the total number of positive examples.

1. High Recall indicates the class is correctly recognized (small number of FN).

2. Low Recall indicates the class is incorrectly recognized (large number of FN).

Recall is given by the relation:

$$\textit{Recall} = \frac{TP}{TP + FN}$$

Precision:
To get the value of precision, we divide the total number of correctly classified positive examples by the total number of predicted positive examples.
1. High Precision indicates an example labeled as positive is indeed positive (a small number of FP).
2. Low Precision indicates an example labeled as positive is indeed positive (large number of FP).

The relation gives precision:

$$\textit{Precision} = \frac{TP}{TP + FP}$$

Remember:-
High recall, low precision: This means that most of the positive examples are correctly recognized (low FN), but there are a lot of false positives.
Low recall, high precision: This shows that we miss a lot of positive examples (high FN), but those we predict as positive are indeed positive (low FP).

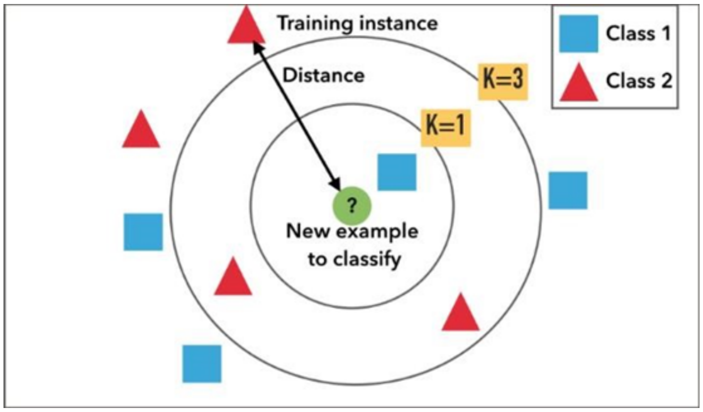
F-measure/F1-Score:
Since we have two measures (Precision and Recall), it helps to have a measurement that represents both of them. We calculate an **F-measure, which uses Harmonic Mean in place of Arithmetic Mean as it punishes the extreme values more.**

The F-Measure will always be nearer to the smaller value of Precision or Recall.

$$\textit{F - measure} = \frac{2*\text{Recall}*\text{Precision}}{\text{Recall} + \text{Precision}}$$

16. **What is KNN Classifier ?**

KNN means K-Nearest Neighbour Algorithm. It can be used for both classification and regression.



It is the simplest machine learning algorithm. Also known as lazy learning (why? Because it does not create a generalized model during the time of training, so the testing phase is very important where it does the actual job. Hence Testing is very costly - in terms of time & money). Also called an instancebased or memory-based learning

In k-NN classification, the output is a class membership. An object is classified by a plurality vote of its neighbors, with the object being assigned to the class most common among its k nearest neighbors (k is a positive integer, typically small). If k = 1, then the object is assigned to the class of that single nearest neighbor.

In k-NN regression, the output is the property value for the object. This value is the average of the values of k nearest neighbors.

Distance functions

Euclidean $$\sqrt{\sum_{i=1}^{k}(x_i - y_i)^2}$$

Manhattan $$\sum_{i=1}^{k}|x_i - y_i|$$

Minkowski $$\left(\sum_{i=1}^{k}(|x_i - y_i|)^q\right)^{1/q}$$

All three distance measures are only valid for continuous variables. In the instance of categorical variables, the Hamming distance must be used.

Hamming Distance

$$D_H = \sum_{i=1}^{k} |x_i - y_i|$$

$$x = y \Rightarrow D = 0$$
$$x \neq y \Rightarrow D = 1$$

X	Y	Distance
Male	Male	0
Male	Female	1

How to choose the value of K: K value is a hyperparameter which needs to choose during the time of model building

Also, a small number of neighbors are most flexible fit, which will have a low bias, but the high variance and a large number of neighbors will have a smoother decision boundary, which means lower variance but higher bias.

We should choose an odd number if the number of classes is even. It is said the most common values are to be 3 & 5.

17. What is the Pipeline in sklearn ?

A pipeline is what chains several steps together, once the initial exploration is done. For example, some codes are meant to transform features—normalize numerically, or turn text into vectors, or fill up missing data, and they are transformers; other codes are meant to predict variables by fitting an algorithm, such as random forest or support vector machine, they are estimators. Pipeline chains all these together, which can then be applied to training data in block.

Example of a pipeline that imputes data with the most frequent value of each column, and then fit a decision tree classifier.

```
From sklearn.pipeline import Pipeline
 steps = [('imputation', Imputer(missing_values='NaN', strategy = 'most_frequent', axis=0)),
         ('clf', DecisionTreeClassifier())]
 pipeline = Pipeline(steps)
 clf = pipeline.fit(X_train,y_train)```
```

Instead of fitting to one model, it can be looped over several models to find the best one.

```
classifiers = [ KNeighborsClassifier(5), RandomForestClassifier(), GradientBoostingClassifier()] for clf in classifiers:
steps = [('imputation', Imputer(missing_values='NaN', strategy = 'most_frequent', axis=0)),
     ('clf', clf)]
```

pipeline = Pipeline(steps)

I also learned the pipeline itself can be used as an estimator and passed to cross-validation or grid search. from sklearn.model_selection import KFold from sklearn.model_selection import cross_val_score kfold = KFold(n_splits=10, random_state=seed)
results = cross_val_score(pipeline, X_train, y_train, cv=kfold) print(results.mean())

18. What is Principal Component Analysis(PCA), and why do we do it?

The main idea of principal component analysis (PCA) is to reduce the dimensionality of a data set consisting of many variables correlated with each other, either heavily or lightly, while retaining the variation present in the dataset, up to the maximum extent. The same is done by transforming the variables to a new set of variables, which are known as the principal components (or simply, the PCs) and are orthogonal, ordered such that the retention of variation present in the original variables decreases as we move down in the order. So, in this way, the 1st principal component retains maximum variation that was present in the original components. The principal components are the eigenvectors of a covariance matrix, and hence they are orthogonal.

Main important points to be considered:

1. Normalize the data
2. Calculate the covariance matrix
3. Calculate the eigenvalues and eigenvectors
4. Choosing components and forming a feature vector
5. Forming Principal Components

19. What do you do if there are outliers?

Following are the approaches to handle the outliers:

1. Drop the outlier records
2. Assign a new value: If an outlier seems to be due to a mistake in your data, you try imputing a value.
3. If percentage-wise the number of outliers is less, but when we see numbers, there are several, then, in that case, dropping them might cause a loss in insight. We should group them in that case and run our analysis separately on them.

20. What are the encoding techniques you have applied with Examples ?

In many practical data science activities, the data set will contain categorical variables. These variables are typically stored as text values". Since machine learning is based on mathematical equations, it would cause a problem when we keep categorical variables as is.

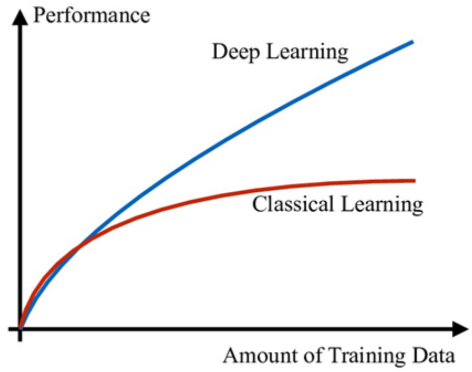
Let's consider the following dataset of fruit names and their weights.
Some of the common encoding techniques are:
Label encoding: In label encoding, we map each category to a number or a label. The labels chosen for the categories have no relationship. So categories that have some ties or are close to each other lose such information after encoding.
One - hot encoding: In this method, we map each category to a vector that contains 1 and 0 denoting the presence of the feature or not. The number of vectors depends on the categories which we want to keep. For high cardinality features, this method produces a lot of columns that slows down the learning significantly.

21. What is the difference between machine learning and deep learning?

Machine Learning | deep learning
Machine Learning is a technique to learn from that data and then apply what has been learnt to make an informed decision | The main difference between deep and machine learning is, machine learning models become better progressively but the model still needs some guidance. If a machine learning model returns an inaccurate prediction then the programmer need s to fix that problem explicitly but in the case of deep learning, the model does it by himself.
>Machine Learning can perform well with small size data also | Deep Learing does not perform as good with smaller datasets.
>Machine learning can work on some low-
end machines also | Deep Learning involves many matrix multiplication operations which are better suited for GPUs
>Features need to be identified and extracted as per the domain before pushing them to the algorithm | Deep learning algorithms try to learn highlevel features from data.
>It is generally recommended to break the problem into smaller chunks, solve them and then combine the results | It generally focusses on solving the problem end to end
>Training time is comparatively less | Training time is comparatively more
>Results are more interpretable | Results Maybe more accurate but less interpretable
> No use of Neural networks | uses neural networks
> Solves comparatively less complex problems | Solves more complex problems.

22. Why deep learning is better than machine learning ?

Though traditional ML algorithms solve a lot of our cases, they are not useful while working with high dimensional data that is where we have a large number of inputs and outputs. For example, in the case of handwriting recognition, we have a large amount of input where we will have different types of inputs associated with different types of handwriting.



The second major challenge is to tell the computer what are the features it should look for that will play an important role in predicting the outcome as well as to achieve better accuracy while doing so.

23. What kind of problem can be solved by using deep learning ?

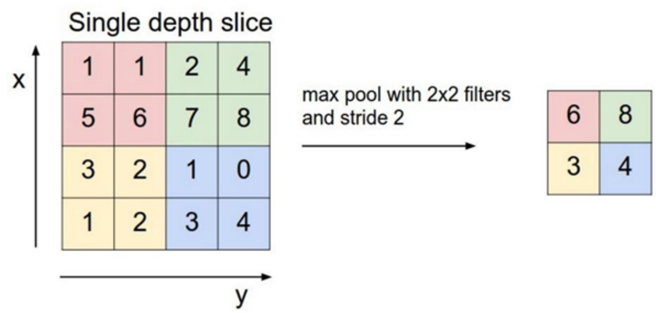
Deep Learning is a branch of Machine Learning, which is used to solve problems in a way that mimics the human way of solving problems. Examples:

- Image recognition
- Object Detection
- Natural Language processing- Translation, Sentence formations, text to speech, speech to text
- understand the semantics of actions

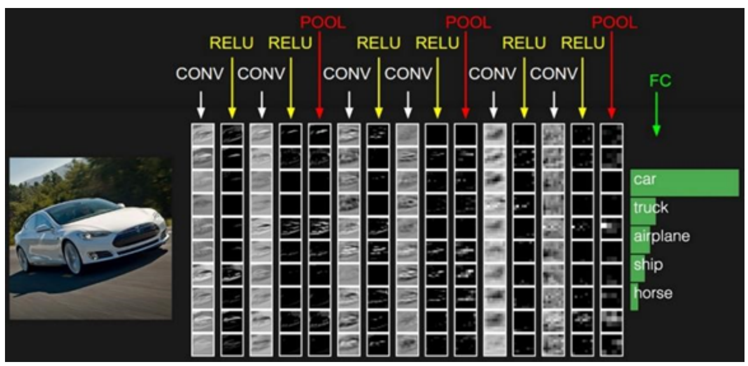
24. What is CNN ?

This is the simple application of a filter to an input that results in inactivation. Repeated application of the same filter to input results in a map of activations called a feature map, indicating the locations and strength of a detected feature in input, such as an image.
Convolutional layers are the major building blocks which are used in convolutional neural networks.
A covnets is the sequence of layers, and every layer transforms one volume to another through differentiable functions. Different types of layers in CNN:
Let's take an example by running a covnets on of image of dimensions 32 x 32 x 3.

1. **Input Layer:** It holds the raw input of image with width 32, height 32 and depth 3.
2. **Convolution Layer:** It computes the output volume by computing dot products between all filters and image patches. Suppose we use a total of 12 filters for this layer we'll get output volume of dimension 32 x 32 x 12.
3. **Activation Function Layer:** This layer will apply the element-wise activation function to the output of the convolution layer. Some activation functions are RELU: max(0, x), Sigmoid: 1/(1+e^-x), Tanh, Leaky RELU, etc. So the volume remains unchanged. Hence output volume will have dimensions 32 x 32 x 12.
4. **Pool Layer:** This layer is periodically inserted within the covnets, and its main function is to reduce the size of volume which makes the computation fast reduces memory and also prevents overfitting. Two common types of pooling layers are max pooling and average pooling. If we use a max pool with 2 x 2 filters and stride 2, the resultant volume will be of dimension 16x16x12.



5. Fully-Connected Layer: This layer is a regular neural network layer that takes input from the previous layer and computes the class scores and outputs the 1-D array of size equal to the number of classes.



25. What is learning Rate ?

Learning Rate

The learning rate controls how much we should adjust the weights concerning the loss gradient. Learning rates are randomly initialised.

Lower the values of the learning rate slower will be the convergence to global minima.

Higher values for the learning rate will not allow the gradient descent to converge

Since our goal is to minimise the function cost to find the optimised value for weights, we run multiples iteration with different weights and calculate the cost to arrive at a minimum cost.

26. What are Epochs?

One Epoch is an ENTIRE dataset is passed forwards and backwards through the neural network.

Since one epoch is too large to feed to the computer at once, we divide it into several smaller batches.

We always use more than one Epoch because one epoch leads to underfitting.

As the number of epochs increases, several times the weight are changed in the neural network and the curve goes from underfitting up to optimal to overfitting curve.

27. List down hyperparameter tuning in deep learning.

The process of setting the hyper-parameters requires expertise and extensive trial and error. There are no simple and easy ways to set hyper-parameters — specifically, learning rate, batch size, momentum, and weight decay.

Approaches to searching for the best configuration:

- Grid Search
- Random Search

Approach

1. Observe and understand the clues available during training by monitoring validation/test loss early in training, tune your architecture and hyper-parameters with short runs of a few epochs.
2. Signs of underfitting or overfitting of the test or validation loss early in the training process are useful for tuning the hyper-parameters.

Tools for Optimizing Hyperparameters

- Sage Maker
- Comet.ml
- Weights &Biases
- Deep Cognition
- Azure ML

28. What do you understand by activation function and error functions?

Error functions

In most learning networks, an error is calculated as the difference between the predicted output and the actual output.

$$J(w) = p - \widehat{p}$$

The function that is used to compute this error is known as Loss Function J(.). Different loss functions will give different errors for the same prediction, and thus have a considerable effect on the performance of the model. One of the most widely used loss function is mean square error, which calculates the square of the difference between the actual values and predicted value. Different loss functions are used to deals with a different type of tasks, i.e. regression and classification.

Regressive loss functions:

Mean Square Error

Absolute error

Smooth Absolute Error

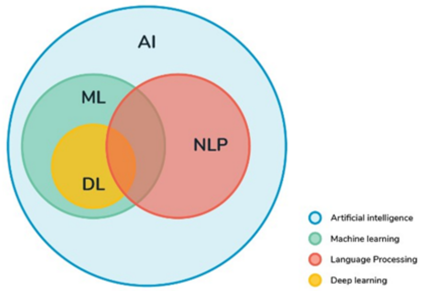
Classification loss functions:

1. Binary Cross-Entropy
2. Negative Log-Likelihood

3. Margin Classifier
4. Soft Margin Classifier

Activation functions decide whether a neuron should be activated or not by calculating a weighted sum and adding bias with it. The purpose of the activation function is to introduce non-linearity into the output of a neuron.

In a neural network, we would update the weights and biases of the neurons based on the error at the outputs. This process is known as back-propagation. Activation function makes the back-propagation possible since the gradients are supplied along with the errors to update the weights and biases.



29. What is NLP ?

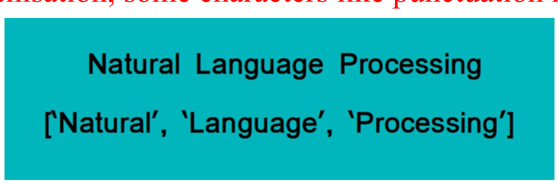
Natural language processing (NLP): It is the branch of artificial intelligence that helps computers understand, interpret and manipulate human language. NLP draws from many disciplines, including computer science and computational linguistics, in its pursuit to fill the gap between human communication and computer understanding.

30. . What are the Libraries we used for NLP?

We usually use these libraries in NLP, which are:
NLTK (Natural language Tool kit), TextBlob, CoreNLP, Polyglot,
Gensim, SpaCy, Scikit-learn
And the new one is Megatron library launched recently.

31. What do you understand by tokenisation?

Tokenisation is the act of breaking a sequence of strings into pieces such as words, keywords, phrases, symbols and other elements called tokens. Tokens can be individual words, phrases or even whole sentences. In the process of tokenisation, some characters like punctuation marks are discarded.



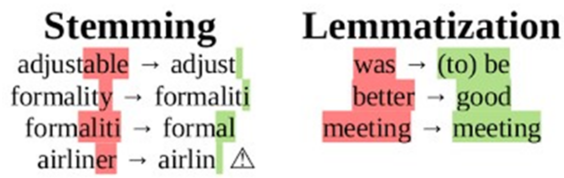
32. What do you understand by stemming?

Stemming: It is the process of reducing inflexions in words to their root forms such as mapping a group of words to the same stem even if stem itself is not a valid word in the Language.

	words	stemmed words
0	connect	connect
1	connected	connect
2	connection	connect
3	connections	connect
4	connects	connect

33. What is lemmatisation?

Lemmatisation: It is the process of the group together the different inflected forms of the word so that they can be analysed as a single item. It is quite similar to stemming, but it brings context to the words. So it links words with similar kind meaning to one word.



34. . What is Bag-of-words model?

We need the way to represent text data for the machine learning algorithms, and the bag-of-words model helps us to achieve the task. This model is very understandable and to implement. It is the way of extracting features from the text for the use in machine learning algorithms.

In this approach, we use the tokenised words for each of observation and find out the frequency of each token.

Let's do an example to understand this concept in depth.

"It is going to rain today."

"Today, I am not going outside."

"I am going to watch the season premiere."

We treat each sentence as the separate document and we make the list of all words from all the three documents excluding the punctuation. We get,

'It', 'is', 'going', 'to', 'rain', 'today' 'I', 'am', 'not', 'outside', 'watch', 'the', 'season', 'premiere.'

The next step is the create vectors. Vectors convert text that can be used by the machine learning algorithm.

We take the first document — "It is going to rain today", and we check the frequency of words from the ten unique words.

"It" = 1

"is" = 1

"going" = 1

"to" = 1

"rain" = 1

"today" = 1

"I" = 0

"am" = 0

"not" = 0

"outside" = 0

Rest of the documents will be:
"It is going to rain today" = [1, 1, 1, 1, 1, 1, 0, 0, 0, 0]
"Today I am not going outside" = [0, 0, 1, 0, 0, 1, 1, 1, 1, 1]
"I am going to watch the season premiere" = [0, 0, 1, 1, 0, 0, 1, 1, 0, 0]
In this approach, each word (a token) is called a "gram". Creating the vocabulary of two-word pairs is called a bigram model.
The process of converting the NLP text into numbers is called vectorisation in ML. There are different ways to convert text into the vectors :
- Counting the number of times that each word appears in the document.
- I am calculating the frequency that each word appears in a document out of all the words in the document.

35. What do you understand by TF-IDF?

TF-IDF: It stands for the term of frequency-inverse document frequency.
TF-IDF weight: It is a statistical measure used to evaluate how important a word is to a document in a collection or corpus. The importance increases proportionally to the number of times a word appears in the document but is offset by the frequency of the word in the corpus.
- **Term Frequency (TF):** is a scoring of the frequency of the word in the current document. Since every document is different in length, it is possible that a term would appear much more times in long documents than shorter ones. The term frequency is often divided by the document length to normalise.

$$TF(t) = \frac{Number\ of\ times\ term\ t\ appears\ in\ a\ document}{Total\ number\ of\ terms\ in\ the\ document}$$

- **Inverse Document Frequency (IDF): It is a scoring of how rare the word is across the documents. It is a measure of how rare a term is, Rarer the term, and more is the IDF score.**

$$IDF(t) = log_e\left(\frac{Total\ number\ of\ documents}{Number\ of\ documents\ with\ term\ t\ in\ it}\right)$$

Thus,

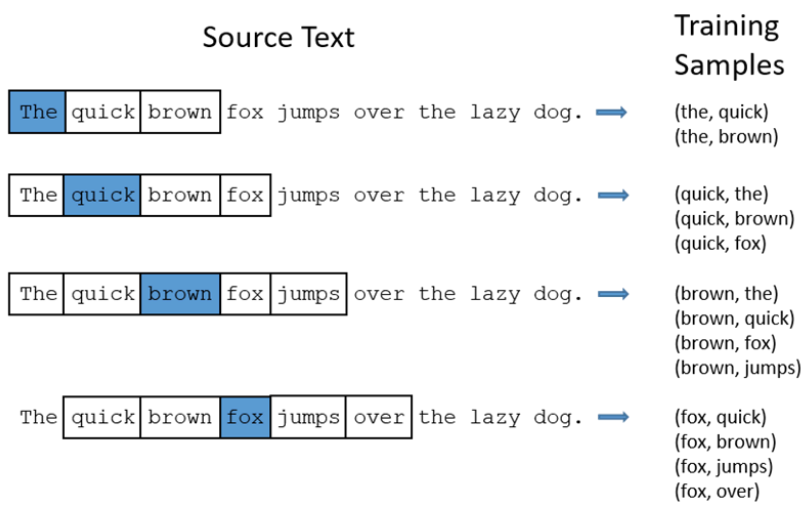
$$TF - IDFscore = TF * IDF$$

36. What is Word2vec?

Word2Vec is a shallow, two-layer neural network which is trained to reconstruct linguistic contexts of words. It takes as its input a large corpus of words and produces a vector space, typically of several of hundred dimensions, with each of unique word in the corpus being assigned to the corresponding vector in space.
Word vectors are positioned in a vector space such that words which share common contexts in the corpus are located close to one another in the space.
Word2Vec is a particularly computationally-efficient predictive model for learning word embeddings from raw text.

Word2Vec is a group of models which helps derive relations between a word and its contextual words. Let's look at two important models inside Word2Vec: Skip-grams and CBOW.

SKIP-gram



In Skip-gram model, we take a centre word and a window of context (neighbour) words, and we try to predict the context of words out to some window size for each centre word. So, our model is going to define a probability distribution, i.e. probability of a word appearing in the context given a centre word and we are going to choose our vector representations to maximise the probability.

Continuous Bag-of-Words (CBOW)

CBOW predicts target words (e.g. 'mat') from the surrounding context words ('the cat sits on the'). Statistically, it affects that CBOW smoothes over a lot of distributional information (by treating an entire context as one observation). For the most part, this turns out to be a useful thing for smaller datasets.

1. I enjoy flying.
2. I like NLP.
3. I like deep learning.

The resulting counts matrix will then be:

$$X = \begin{array}{c|cccccccc} & I & like & enjoy & deep & learning & NLP & flying & . \\ \hline I & 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ like & 2 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ enjoy & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ deep & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ learning & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ NLP & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ flying & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ . & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{array}$$

This was about converting words into vectors. But where does the "learning" happen? Essentially, we begin with small random initialisation of word vectors. Our predictive model learns the vectors by minimising the loss function. In Word2vec, this happens with feedforward neural networks and optimisation techniques such as Stochastic gradient descent. There are also count-based models which make the co-occurrence count matrix of the words in our corpus; we have a very large matrix with each row for the "words" and columns for the "context". The number of "contexts" is, of course very large, since it is very essentially combinatorial in size. To overcome this issue, we apply SVD to a matrix. This reduces the dimensions of the matrix to retain maximum pieces of information.

37. . What is Time-Series forecasting?

Time series forecasting is a technique for the prediction of events through a sequence of time. The technique is used across many fields of study, from the geology to behaviour to economics. The techniques predict future events by analysing the trends of the past, on the assumption that future trends will hold similar to historical trends.

38. What is the difference between in Time series and regression?

Time-series:

1. Whenever data is recorded at regular intervals of time.
2. Time-series forecast is Extrapolation.
3. Time-series refers to an ordered series of data.

Regression:

1. Whereas in regression, whether data is recorded at regular or irregular intervals of time, we can apply
2. Regression is Interpolation.
3. Regression refer both ordered and unordered series of data.

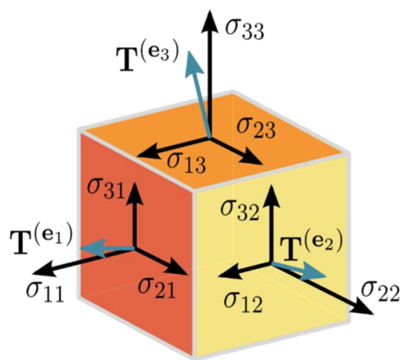
39. What is Tensorflow ?

TensorFlow: TensorFlow is an open-source software library released in 2015 by Google to make it easier for the developers to design, build, and train deep learning models. TensorFlow is originated as an internal library that the Google developers used to build the models in house, and we expect additional functionality to be added in the open-source version as they are tested and vetted in internal flavour. Although TensorFlow is the only one of several options available to the developers and we choose to use it here because of thoughtful design and ease of use.

At a high level, TensorFlow is a Python library that allows users to express arbitrary computation as a graph of data flows. Nodes in this graph represent mathematical operations, whereas edges represent data that is communicated from one node to another. Data in TensorFlow are represented as tensors, which are multidimensional arrays. Although this framework for thinking about computation is valuable in many different fields, TensorFlow is primarily used for deep learning in practice and research.

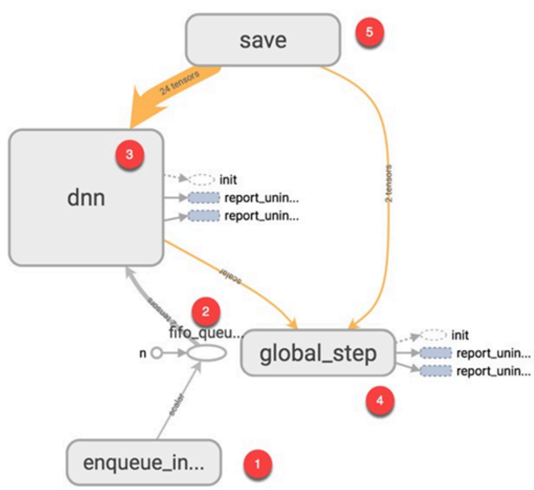
40. What is Tensora ?

Tensor: In mathematics, it is an algebraic object that describes the linear mapping from one set of algebraic objects to the another. Objects that the tensors may map between include, but are not limited to the vectors, scalars and recursively, even other tensors (for example, a matrix is the map between vectors and thus a tensor. Therefore the linear map between matrices is also the tensor). Tensors are inherently related to the vector spaces and their dual spaces and can take several different forms. For example, a scalar, a vector, a dual vector at a point, or a multi-linear map between vector spaces. Euclidean vectors and scalars are simple tensors. While tensors are defined as independent of any basis. The literature on physics, often referred by their components on a basis related to a particular coordinate system.



41. What is TensorBoard ?

TensorBoard, a suit of visualising tools, is an easy solution to Tensorflow offered by the creators that lets you visualise the graphs, plot quantitative metrics about the graph with additional data like images to pass through it.



This one is some example of how the TensorBoard is working.

42. What are the features of Tensorflow ?

- One of the main features of TensorFlow is its ability to build neural networks.
- By using these neural networks, machines can perform logical thinking and learn similar to humans.

- There are the other tensors for processing, such as data loading, preprocessing, calculation, state and outputs.
- It considered not only as deep learning but also as the library for performing the tensor calculations, and it is the most excellent library when considered as the deep learning framework that can also describe basic calculation processing.
- TensorFlow describes all calculation processes by calculation graph, no matter how simple the calculation is.

43. What are the advantages of TensorFlow?

- It allows Deep Learning.
- It is open-source and free.
- It is reliable (and without major bugs) • It is backed by Google and a good community.
- It is a skill recognised by many employers.
- It is easy to implement.

44. List a few limitations of Tensorflow.

- Has the GPU memory conflicts with Theano if imported in the same scope.
- It has dependencies with other libraries.
- Requires prior knowledge of the advanced calculus and linear algebra along with the pretty good understanding of machine learning.

45. What are the cases of Tensorflow ?

Tensorflow is an important tool of deep learning, it has mainly five use cases, and they are:

- Time Series
- Image recognition
- Sound Recognition
- Video detection
- Text-based Applications

46. What are the very important steps of Tensorflow architecture?

There are three main steps in the Tensorflow architecture are:

- Pre-process the Data
- Build a Model
- Train and estimate the model

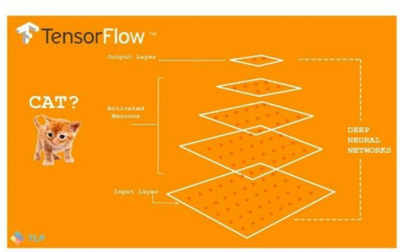
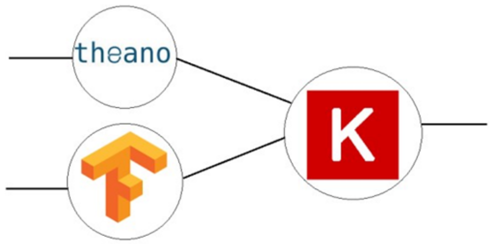


Image Recognition
Classification using Softmax Regressions and Convolutional Neural Networks

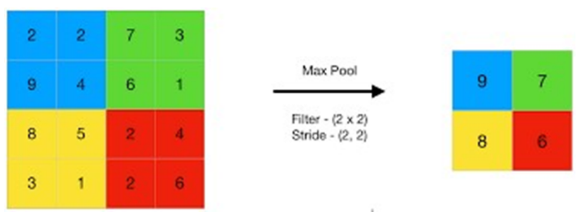
47. What is Keras ?

Keras: It is an Open Source Neural Network library written in Python that runs on the top of Theano or Tensorflow. It is designed to be the modular, fast and easy to use. It was developed by François Chollet, a Google engineer.



48. What is pooling layer ?

Pooling layer: It is generally used in reducing the spatial dimensions and not depth, on a convolutional neural network model.



49. What is the difference between CNN and RNN?

CNN (Convolutional Neural Network)

- Best suited for spatial data like images
- CNN is powerful compared to RNN
- This network takes a fixed type of inputs and outputs
- These are the ideal for video and image processing

RNN (Recurrent Neural Network)

- Best suited for sequential data
- RNN supports less feature set than CNN.
- This network can manage the arbitrary input and output lengths.
- It is ideal for text and speech analysis.

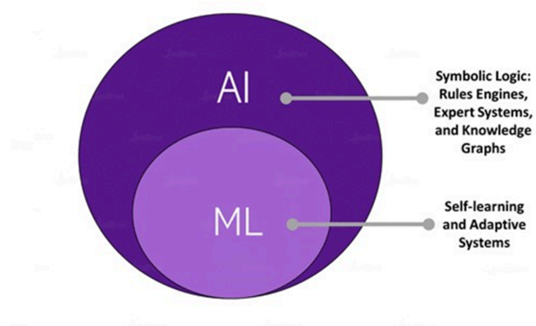
50. What are the benefits of Tensorflow over other libraries?

The following benefits are:

- Scalability
- Visualisation of Data
- Debugging facility
- Pipelining

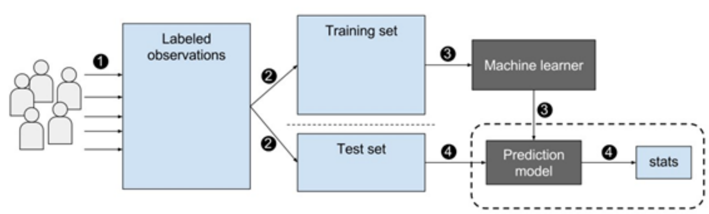
51. How would you define Machine Learning?

Machine learning: It is an application of artificial intelligence (AI) that provides systems the ability to learn automatically and to improve from experiences without being programmed. It focuses on the development of computer applications that can access the data and used it to learn for themselves.
The process of learning starts with the observations or data, such as examples, direct experience, or instruction, to look for the patterns in data and to make better decisions in the future based on examples that we provide. The primary aim is to allow the computers to learn automatically without human intervention or assistance and adjust actions accordingly.



52. What is a labeled training set?

Machine learning is derived from the availability of the labeled data in the form of a **training set and test set** that is used by the learning algorithm. The separation of data into the training portion and a test portion is the way the algorithm learns. We split up the data containing known response variable values into two pieces. The training set is used to train the algorithm, and then you use the trained model on the test set to predict the variable response values that are already known. The final step is to compare with the predicted responses against actual (observed) responses to see how close they are. The difference is the test error metric. Depending on the test error, you can go back to refine the model and repeat the process until you're satisfied with the accuracy.

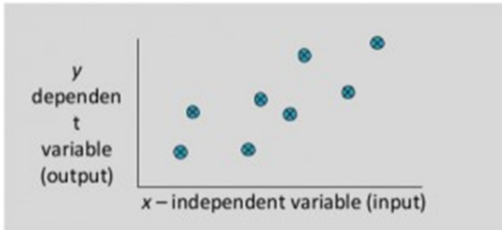


53. What are the two common supervised tasks?

The two common supervised tasks are regression and classification.

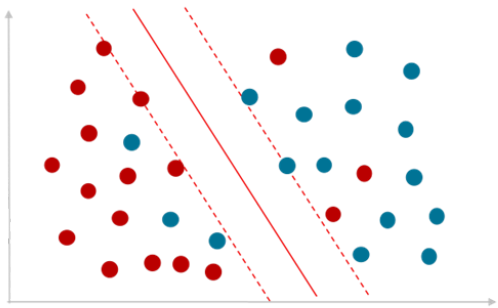
Regression-

The regression problem is when the output variable is the real or continuous value, such as "salary" or "weight." Many different models can be used, and the simplest is linear regression. It tries to fit the data with the best hyper-plane, which goes through the points.



Classification

It is the type of supervised learning. It specifies the class to which the data elements belong to and is best used when the output has finite and discrete values. It predicts a class for an input variable, as well.

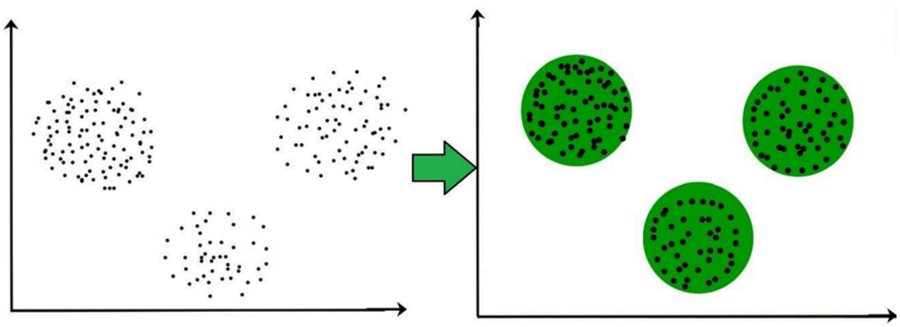


54. Can you name four common unsupervised tasks?

The common unsupervised tasks include clustering, visualization, dimensionality reduction, and association rule learning.

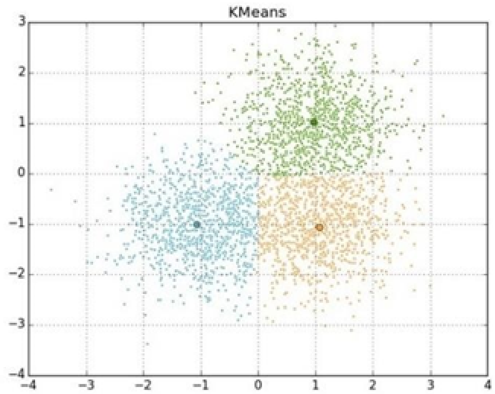
Clustering

It is a Machine Learning technique that involves the grouping of the data points. Given a set of data points, and we can use a clustering algorithm to classify each data point into the specific group. In theory, data points that lie in the same group should have similar properties and/ or features, and data points in the different groups should have high dissimilar properties and/or features. Clustering is the method of unsupervised learning and is a common technique for statistical data analysis used in many fields.



Visualization

Data visualization is the technique that uses an array of static and interactive visuals within the specific context to help people to understand and make sense of the large amounts of data. The data is often displayed in the story format that visualizes patterns, trends, and correlations that may go otherwise unnoticed. It is regularly used as an avenue to monetize data as the product. An example of using monetization and data visualization is Uber. The app combines visualization with real-time data so that customers can request a ride.

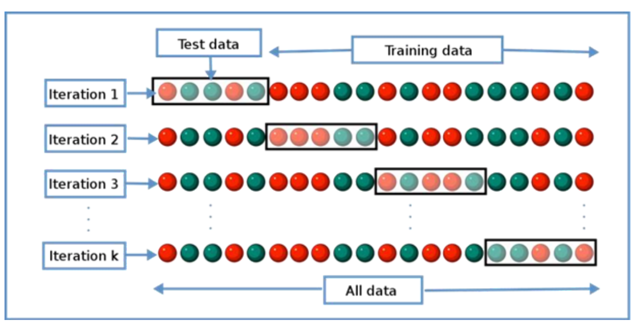


55. What is cross-validation?

Cross-validation: It is a technique for evaluating Machine Learning models by training several Machine Learning models on subsets of available input data and evaluating them on the complementary subset of data. Use cross-validation to detect overfitting, i.e., failing to generalize a pattern.

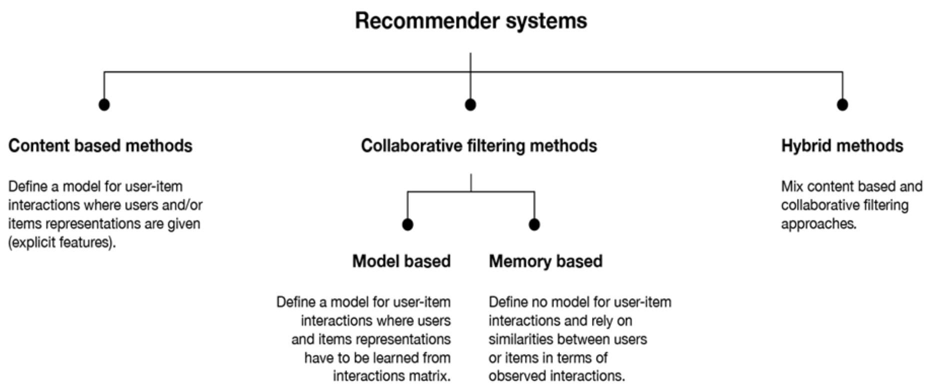
There are three steps involved in cross-validation are as follows :

- Reserve some portion of the sample dataset.
- Using the rest dataset and train models.
- Test the model using a reserve portion of the data-set.



56. What is a Recommender System?

A recommender system is today widely deployed in multiple fields like movie recommendations, music preferences, social tags, research articles, search queries and so on. The recommender systems work as per collaborative and content-based filtering or by deploying a personality-based approach. This type of system works based on a person's past behavior in order to build a model for the future. This will predict the future product buying, movie viewing or book reading by people. It also creates a filtering approach using the discrete characteristics of items while recommending additional items.



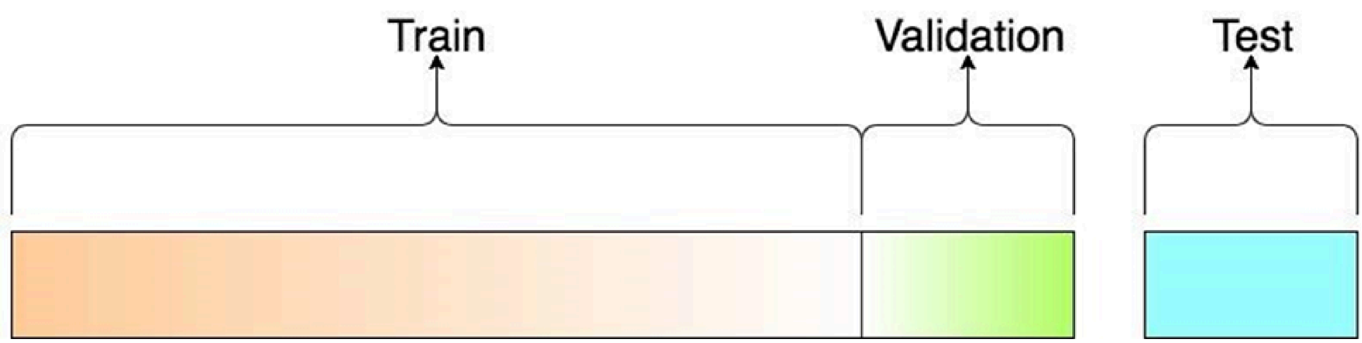
57. Explain the difference between a Test Set and a Validation Set?

Validation set can be considered as a part of the training set as it is used for parameter selection and to avoid Overfitting of the model being built. On the other hand, test set is used for testing or evaluating the performance of a trained machine leaning model. In simple terms ,the differences can be summarized as-

Training Set is to fit the parameters i.e. weights.

Test Set is to assess the performance of the model i.e. evaluating the predictive power and generalization.

Validation set is to tune the parameters.



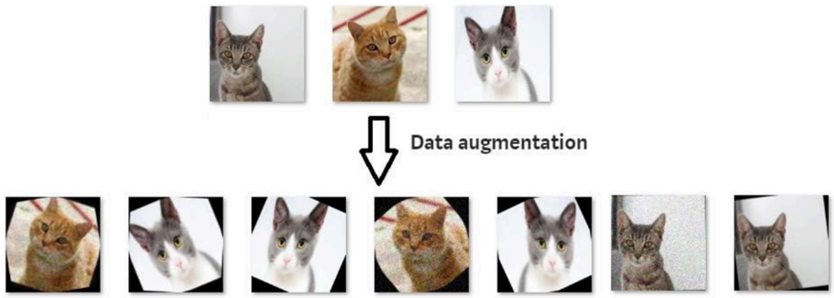
58. What is data augmentation? Can you give some examples?

Data augmentation is a technique for synthesizing new data by modifying existing data in such a way that the target is not changed, or it is changed in a known way.

Computer vision is one of fields where data augmentation is very useful. There are many modifications that we can do to images:

- Resize
- Horizontal or vertical flip
- Rotate
- Add noise
- Deform
- Modify colors

Each problem needs a customized data augmentation pipeline. For example, on OCR, doing flips will change the text and won't be beneficial; however, resizes and small rotations may help.



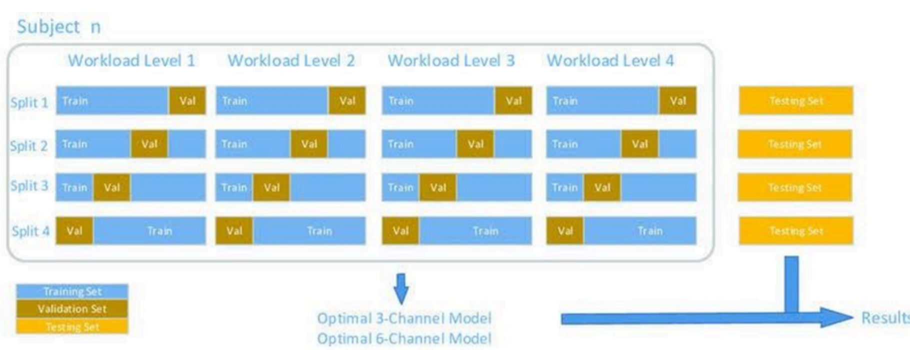
59. What is stratified cross-validation and when should we use it?

Cross-validation is a technique for dividing data between training and validation sets. On typical crossvalidation this split is done randomly. But in stratified cross-validation, the split preserves the ratio of the categories on both the training and validation datasets.

For example, if we have a dataset with 10% of category A and 90% of category B, and we use stratified cross-validation, we will have the same proportions in training and validation. In contrast, if we use simple cross-validation, in the worst case we may find that there are no samples of category A in the validation set.

Stratified cross-validation may be applied in the following scenarios:

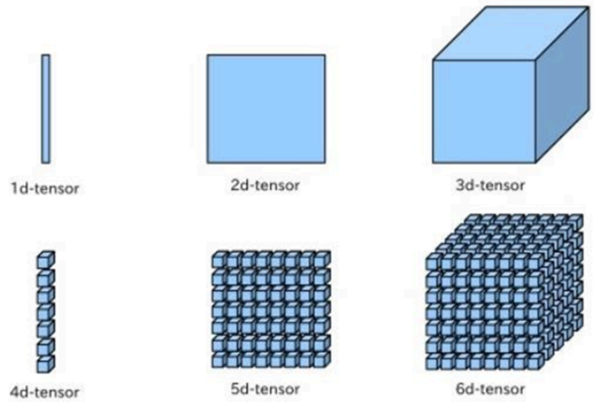
- On a dataset with multiple categories. The smaller the dataset and the more imbalanced the categories, the more important it will be to use stratified cross-validation.
- On a dataset with data of different distributions. For example, in a dataset for autonomous driving, we may have images taken during the day and at night. If we do not ensure that both types are present in training and validation, we will have generalization problems.



60. What are tensors?

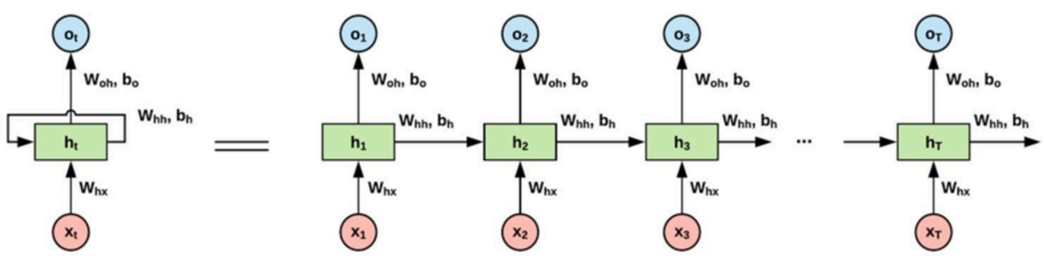
The tensors are no more than a method of presenting the data in deep learning. If put in the simple term, tensors are just multidimensional arrays that allow developers to represent the data in a layer, which means deep learning you are using contains high-level data sets where each dimension represents a different feature.

The foremost benefit of using tensors is it provides the much-needed platform-flexibility and is easy to trainable on CPU. Apart from this, tensors have the auto differentiation capabilities, advanced support system for queues, threads, and asynchronous computation. All these features also make it customizable.



61. **Define the concept of RNN?**

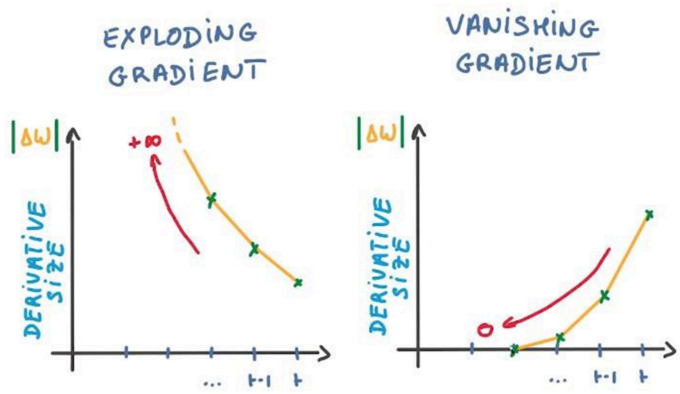
RNN is the artificial neutral which were created to analyze and recognize the patterns in the sequences of the data. Due to their internal memory, RNN can certainly remember the things about the inputs they receive.



Most common issues faced with RNN

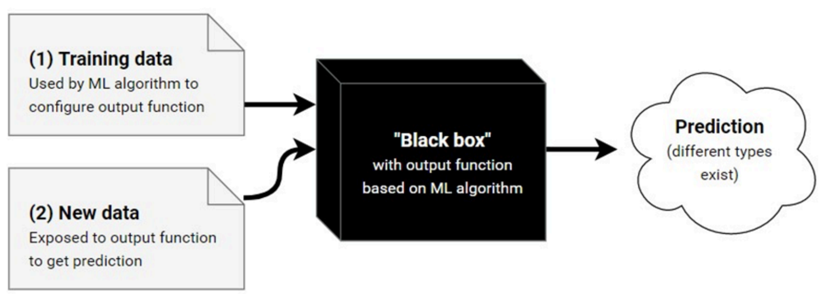
Although RNN is around for a while and uses backpropagation, there are some common issues faced by developers who work it. Out of all, some of the most common issues are:

- Exploding gradients
- Vanishing gradients



62. Why are deep learning models referred as black boxes?

Lately, the concept of deep learning being a black box has been floating around. A black box is a system whose functioning cannot be properly grasped, but the output produced can be understood and utilized. Now, since most models are mathematically sound and are created based on legit equations, how is it possible that we do not know how the system works?



First, it is almost impossible to visualize the functions that are generated by a system. Most machine learning models end up with such complex output that a human can't make sense of it.

Second, there are networks with millions of hyperparameters. As a human, we can grasp around 10 to 15 parameters. But analysing a million of them seems out of the question.

Third and most important, it becomes very hard, if not impossible, to trace back why the system made the decisions it did. This may not sound like a huge problem to worry about but consider the case of a self driving car. If the car hits someone on the road, we need to understand why that happened and prevent it. But this isn't possible if we do not understand how the system works.

To make a deep learning model not be a black box, a new field called Explainable Artificial Intelligence or simply, Explainable AI is emerging. This field aims to be able to create intermediate results and trace back the decision-making process of a system.

63. Where is the confusion matrix used? Which module would you use to show it?

In machine learning, confusion matrix is one of the easiest ways to summarize the performance of your algorithm.

At times, it is difficult to judge the accuracy of a model by just looking at the accuracy because of problems like unequal distribution. So, a better way to check how good your model is, is to use a confusion matrix.

First, let’s look at some key terms.
Classification accuracy – This is the ratio of the number of correct predictions to the number of predictions made

True positives – Correct predictions of true events
False positives – Incorrect predictions of true events
True negatives – Correct predictions of false events
False negatives – Incorrect predictions of false events.

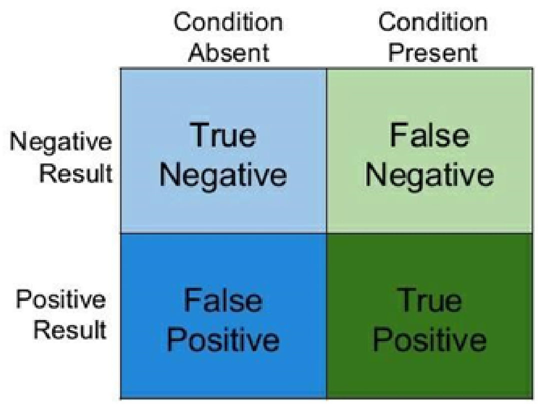
The confusion matrix is now simply a matrix containing true positives, false positives, true negatives, false negatives.

		Predicted Class		
		Positive	**Negative**	
Actual Class	**Positive**	True Positive (TP)	False Negative (FN) **Type II Error**	**Sensitivity** $\frac{TP}{(TP+FN)}$
	Negative	False Positive (FP) **Type I Error**	True Negative (TN)	**Specificity** $\frac{TN}{(TN+FP)}$
		Precision $\frac{TP}{(TP+FP)}$	**Negative Predictive Value** $\frac{TN}{(TN+FN)}$	**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$

64. Accuracy?

It is the most intuitive performance measure and it simply a ratio of correctly predicted to the total observations. We can say as, if we have high accuracy, then our model is best. Yes, we could say that accuracy is a great measure but only when you have symmetric datasets where false positives and false negatives are almost same.

Accuracy = True Positive + True Negative / (True Positive +False Positive + False Negative + True Negative)



65. What is Precision?

It is also called as the positive predictive value. Number of correct positives in your model that predicts compared to the total number of positives it predicts.

Precision = True Positives / (True Positives + False Positives) Precision = True Positives / Total predicted positive

It is the number of positive elements predicted properly divided by the total number of positive elements predicted.

We can say Precision is a measure of exactness, quality, or accuracy. High precision Means that more or all of the positive results you predicted are correct.

66. What is Recall ?

Recall we can also called as sensitivity or true positive rate.

It is several positives that our model predicts compared to the actual number of positives in our data.

Recall = True Positives / (True Positives + False Positives) Recall = True Positives / Total Actual Positive

Recall is a measure of completeness. High recall which means that our model classified most or all of the possible positive elements as positive.

67. : What is F1 Score?

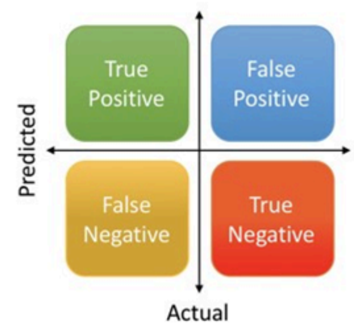
We use Precision and recall together because they complement each other in how they describe the effectiveness of a model. The F1 score that combines these two as the weighted harmonic mean of precision and recall.

F1 Score = 2 * (Precision * Recall) / (Precision + Recall)

$$\textbf{Precision} = \frac{\text{True Positive}}{\text{Actual Results}} \quad \text{or} \quad \frac{\text{True Positive}}{\text{True Positive + False Positive}}$$

$$\textbf{Recall} = \frac{\text{True Positive}}{\text{Predicted Results}} \quad \text{or} \quad \frac{\text{True Positive}}{\text{True Positive + False Negative}}$$

$$\textbf{Accuracy} = \frac{\text{True Positive + True Negative}}{\text{Total}}$$



68. What is Bias and Variance trade-off ?

Bias

Bias means it's how far are the predict values from the actual values. If the average predicted values are far off from the actual values, then we called as this one have high bias.
When our model has a high bias, then it means that our model is too simple and does not capture the complexity of data, thus underfitting the data.

Variance

It occurs when our model performs good on the trained dataset but does not do well on a dataset that it is not trained on, like a test dataset or validation dataset. It tells us that actual value is how much scattered from the predicted value.

Because of High variance it cause overfitting that implies that the algorithm models random noise present in the training data.
When model have high variance, then model becomes very flexible and tune itself to the data points of the training set.

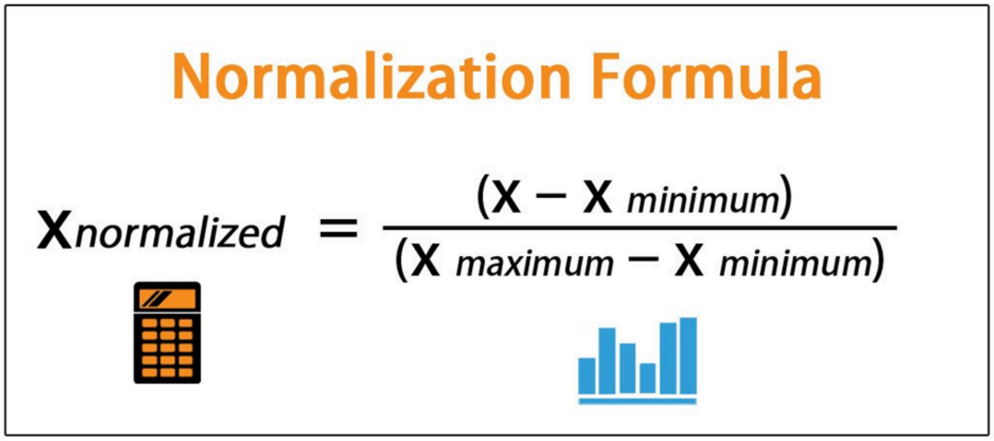
69. Why is normalization required before applying any machine learning model? What module can you use to perform normalization?

Normalization is a process that is required when an algorithm uses something like distance measures. Examples would be clustering data, finding cosine similarities, creating recommender systems.

Normalization is not always required and is done to prevent variables that are on higher scale from affecting outcomes that are on lower levels. For example, consider a dataset of employees' income. This data won't be on the same scale if you try to cluster it. Hence, we would have to normalize the data to prevent incorrect clustering.

A key point to note is that normalization does not distort the differences in the range of values.
A problem we might face if we don't normalize data is that gradients would take a very long time to descend and reach the global maxima/ minima.

For numerical data, normalization is generally done between the range of 0 to 1.
The general formula is:
Xnew = (x-xmin)/(xmax-xmin)



70. . What is the difference between feature selection and feature extraction?

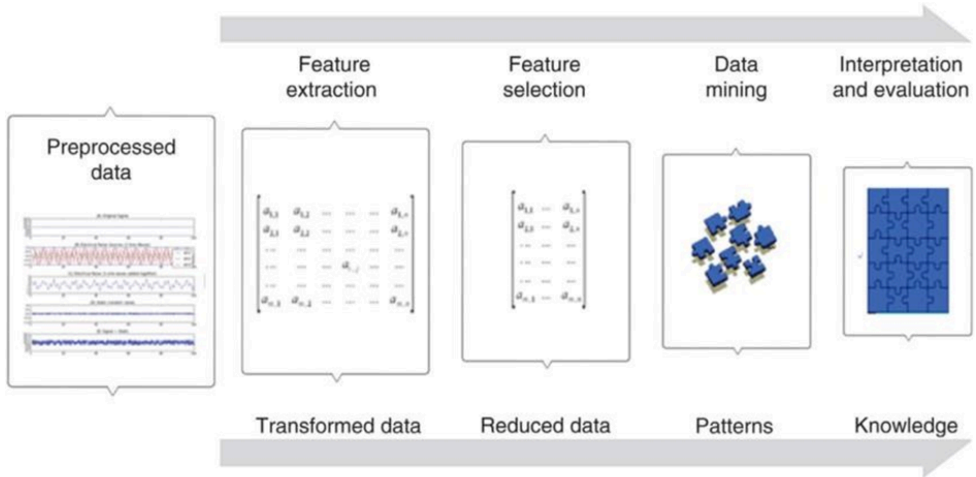
Feature selection and feature extraction are two major ways of fixing the curse of dimensionality

1. Feature selection:

Feature selection is used to filter a subset of input variables on which the attention should focus.
Every other variable is ignored. This is something which we, as humans, tend to do subconsciously.
Many domains have tens of thousands of variables out of which most are irrelevant and redundant.
Feature selection limits the training data and reduces the amount of computational resources used. It can significantly improve a learning algorithms performance.
In summary, we can say that the goal of feature selection is to find out an optimal feature subset. This might not be entirely accurate, however, methods of understanding the importance of features also exist. Some modules in python such as Xgboost help achieve the same.

2. Feature extraction

Feature extraction involves transformation of features so that we can extract features to improve the process of feature selection. For example, in an unsupervised learning problem, the extraction of bigrams from a text, or the extraction of contours from an image are examples of feature extraction.
The general workflow involves applying feature extraction on given data to extract features and then apply feature selection with respect to the target variable to select a subset of data. In effect, this helps improve the accuracy of a model.



71. What is Text Generation ?

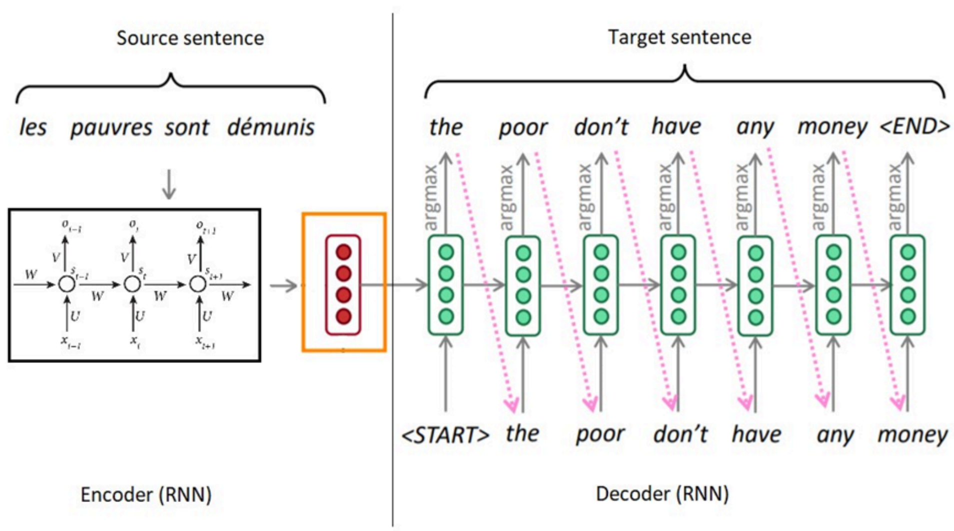
Text Generation: It is a type of the Language Modelling problem. Language Modelling is the core problem for several of natural language processing tasks such as speech to text, conversational system, and the text summarization. The trained language model learns the likelihood of occurrence of the word based on the previous sequence of words used in the text. Language models can be operated at the character level, n-gram level, sentence level or even paragraph level.

A language model is at the core of many NLP tasks, and is simply a probability distribution over a sequence of words:

$$p(w_1, \ldots, w_n)$$

It can also be used to estimate the conditional probability of the next word in a sequence:

$$p(w_n \mid w_1, \ldots, w_{n-1})$$



72. What is Text Summarization?

We all interact with the applications which uses the text summarization. Many of the applications are for the platform which publishes articles on the daily news, entertainment, sports. With our busy schedule, we like to read the summary of those articles before we decide to jump in for reading entire article. Reading a summary helps us to identify the interest area, gives a brief context of the story.

Text summarization is a subdomain of Natural Language Processing (NLP) that deals with extracting summaries from huge chunks of texts. There are two main types of techniques used for text summarization: NLP-based techniques and deep learning-based techniques.

Text summarization: It refers to the technique of shortening long pieces of text. The intention is to create the coherent and fluent summary having only the main points outlined in the document.
How text summarization works:

The two types of summarization, abstractive and the extractive summarization.

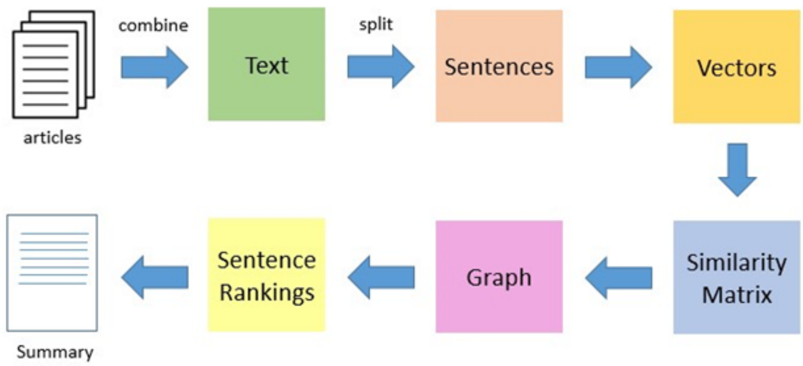
1. **Abstractive Summarization:** It select words based on the semantic understanding; even those words did not appear in the source documents. It aims at producing important material in the new way. They interprets and examines the text using advanced natural language techniques to generate the new shorter text that conveys the most critical information from the original text.
It can be correlated in the way human reads the text article or blog post and then summarizes in their word.

Input document → understand context → semantics → create own summary.

2. **Extractive Summarization:** It attempt to summarize articles by selecting the subset of words that retain the most important points.

This approach weights the most important part of sentences and uses the same to form the summary. Different algorithm and the techniques are used to define the weights for the sentences and further rank them based on importance and similarity among each other.

Input document → sentences similarity → weight sentences → select sentences with higher rank.

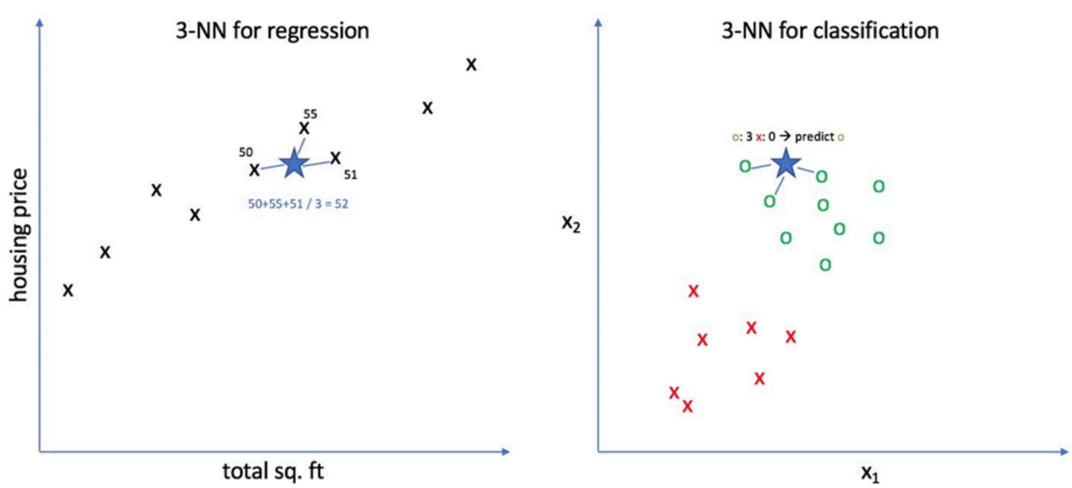


73. . Explain the differences between KNN classifier and KNN regression methods.

They are quite similar. Given a value for KK and a prediction point x0x0, KNN regression first identifies the KK training observations that are closes to x0x0, represented by N0. It then estimates f(x0) using the average of all the training responses in N0. In other words,

$$\hat{f}(x_0) = \frac{1}{K} \sum_{x_i \in N_0} y_i$$

So the main difference is the fact that for the classifier approach, the algorithm assumes the outcome as the class of more presence, and on the regression approach, the response is the average value of the nearest neighbors.



74. What is correlation and the covariance in the statistics?

The Covariance and Correlation are two mathematical concepts; these two approaches are widely used in the statistics. Both Correlation and the Covariance establish the relationship and also measures the dependency between the two random variables, the work is similar between these two, in the mathematical terms, they are different from each other.

Correlation: It is the statistical technique that can show whether and how strongly pairs of variables are related. For example, height and weight are related; taller people tend to be heavier than shorter people. The relationship isn't perfect. People of the same height vary in weight, and you can easily think of two people you know where the shorter one is heavier than the taller one. Nonetheless, the average weight of people 5'5'' is less than the average weight of people 5'6'', and their average weight is less than that of people 5'7'', etc. Correlation can tell you just how much of the variation in peoples' weights is related to their heights.

Correlation Formula

$$\rho_{xy} = \frac{Con(r_x, r_y)}{\sigma_x \sigma_y}$$

x It measures the directional relationship between the returns on two assets. The positive covariance means that asset returns move together while a negative covariance means they move inversely. Covariance is calculated by analyzing at-return surprises (standard deviations from the expected return) or by multiplying the correlation between the two variables by the standard deviation of each variable.

Covariance Formula

For Population

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{N}$$

For Sample

$$Cov(x,y) = \frac{\Sigma (x_i - \bar{x}) * (y_i - \bar{y})}{(N - 1)}$$

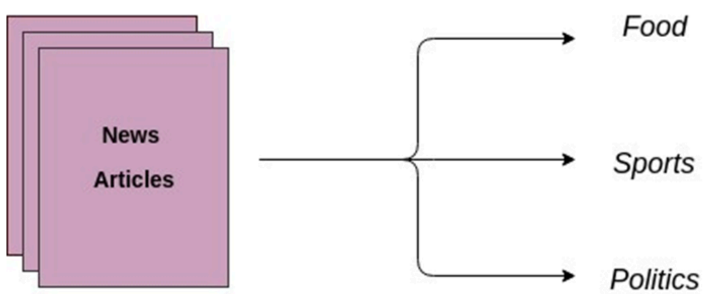
75. What is Text classification in NLP?

Text classification is also known as text tagging or text categorization is a process of categorizing text into organized groups. By using NLP, text classification can automatically analyze text and then assign a set of pre-defined tags or categories based on content.

Unstructured text is everywhere on the internet, such as emails, chat conversations, websites, and the social media but it's hard to extract value from given data unless it's organized in a certain way. Doing so used to be a difficult and expensive process since it required spending time and resources to manually sort the data or creating handcrafted rules that are difficult to maintain. Text classifiers with NLP have proven to be a great alternative to structure textual data in a fast, cost-effective, and scalable way.

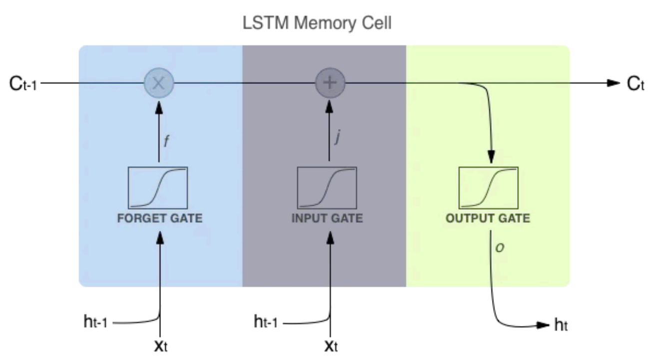
Text classification is becoming an increasingly important part of businesses as it allows us to get insights from data and automate business processes quickly. Some of the most common examples and the use cases for automatic text classification include the following:

- Sentiment Analysis: It is the process of understanding if a given text is talking positively or negatively about a given subject (e.g. for brand monitoring purposes).
- Topic Detection: In this, the task of identifying the theme or topic of a piece of text (e.g. know if a product review is about Ease of Use, Customer Support, or Pricing when analysing customer feedback).
- Language Detection: the procedure of detecting the language of a given text (e.g. know if an incoming support ticket is written in English or Spanish for automatically routing tickets to the appropriate team).

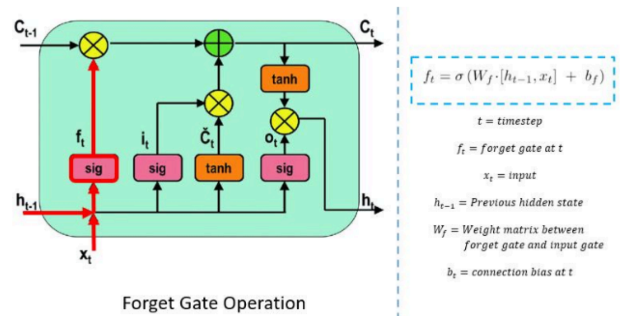


76. Explain LSTM gates ?

In the context of neural networks, particularly Long Short-Term Memory (LSTM) networks, gates are mechanisms that control the flow of information. They help LSTM networks remember long-term dependencies and forget irrelevant information. Here's a brief overview of the three types of gates:



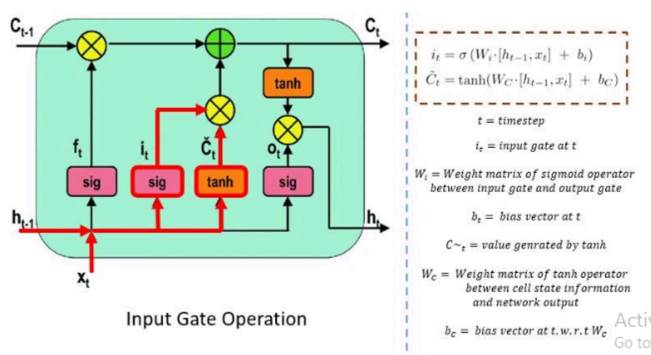
1. **Forget Gate**
   a. **Purpose**: Decides what information from the cell state should be discarded or kept.
   b. **Function:** It takes the previous output and the current input and outputs values between 0 and 1. These values are used to scale the cell state, where values close to 0 mean "forget this information" and values close to 1 mean "keep this information."
   c. **Mathematical Operation**
      i. $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$



Here, $\mathbf{f_t}$ is the forget gate output, $\mathbf{W_f}$ is the weight matrix, $\mathbf{h_{t-1}}$ is the previous output, $\mathbf{x_t}$ is the current input, and $\mathbf{b_f}$ is the bias term. The function σ denotes the sigmoid activation function.

2. **Input Gate**
    a. **Purpose:** Controls how much of the new information from the input should be added to the cell state.
    b. **Function:** It has two components. First, it determines which values to update (using the sigmoid function), and second, it creates a new candidate vector (using the tanh function) that contains the potential updates.
    c. **Mathematical Operation**
        - **it=σ(W_i · [h_{t-1},xt]+b_i)**
        - **$C^{\sim}_t$=tanh(W_C · [h_t–1,x_t]+b_C)**

Input Gate

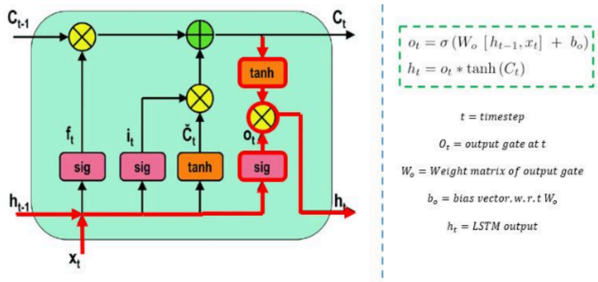


Here, i_t is the input gate output, $C\sim_t$ is the candidate cell state, W_i and $W_C$are weight matrices, and b_i and b_C are bias terms.

3. **Output Gate**
    a. **Purpose:** Decides what part of the cell state should be outputted and sent to the next layer or the next time step.
    b. **Function:** It determines how much of the cell state should be exposed as the output by applying the sigmoid function to decide which parts to output and then applying the tanh function to the cell state to scale it.
    c. **Mathematical Operation:**

$$h_t = \tanh(W_h \cdot [h_{t-1}, x_t] + b_h)$$

i.
$$y_t = W_y \cdot h_t + b_y$$



Here, o_t is the output gate, h_t is the output at the current time step, C_t is the cell state, W_o is the weight matrix for the output gate, and b_o is the bias term.

The forget gate controls what information to discard, the input gate manages what new information to add, and the output gate determines what part of the cell state should be outputted. Together, these gates help LSTM networks handle long-term dependencies effectively.

77. Explain each type of Recurrent Neural Network (RNN)?

1. **Simple RNN**
   a. **Purpose:** Handles sequences by maintaining a hidden state that captures information from previous time steps.
   b. **Structure:** At each time step, the RNN processes an input and updates its hidden state based on both the current input and the previous hidden state. The hidden state is passed to the next time step, allowing the network to maintain a form of memory.
   c. **Mathematical Operation:**

$$h_t = \tanh(W_h \cdot [h_{t-1}, x_t] + b_h)$$

$$y_t = W_y \cdot h_t + b_y$$

Here, h_t is the hidden state at time t, x_t is the input at time t_t, W_h and W_y are weight matrices, and b_h and b_y are biases.

   d. **Limitation:** Simple RNNs struggle with long-term dependencies due to issues like vanishing and exploding gradients.

2. **LSTM RNN (Long Short-Term Memory)**
   a. **Purpose:** Addresses the limitations of simple RNNs, specifically the issues with long-term dependencies.
   b. **Structure:** Uses a more complex architecture with gates (forget gate, input gate, and output gate) to manage the flow of information and maintain long-term memory. LSTMs have a

cell state that runs through the network with minimal changes, which helps in preserving long-term dependencies.

c. **Mathematical Operations:**

- **Forget Gate:** Decides what to forget from the cell state.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

- **Input Gate:** Decides what new information to add to the cell state.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

- **Output Gate:** Decides what to output based on the cell state.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \cdot \tanh(C_t)$$

d. **Advantage : LSTM** are effective at learning long- term dependencies and handling sequence with varying lengenths.

3. **Bidirectional RNN**

a. **Purpose**: Improves the performance of RNNs by processing sequences in both forward and backward directions.

b. **Structure**: Consists of two separate RNNs: one processes the sequence from start to end (forward RNN), and the other processes it from end to start (backward RNN). The outputs from both RNNs are typically concatenated or combined.

c. **Mathematical Operation**:

- Forward RNN:

$$h_t^{\text{forward}} = \tanh(W_h^{\text{forward}} \cdot [h_{t-1}^{\text{forward}}, x_t] + b_h^{\text{forward}})$$

- Backward RNN:

$$h_t^{\text{backward}} = \tanh(W_h^{\text{backward}} \cdot [h_{t+1}^{\text{backward}}, x_t] + b_h^{\text{backward}})$$

- Combined Output:

$$h_t = \text{concat}(h_t^{\text{forward}}, h_t^{\text{backward}})$$

d. **Advantage :** Bidirectional RNNs can capture context from both past and future time steps, which can be beneficial for tasks where understanding both directions in the sequence is important (e.g., text processing).

78. What are the Max Pooling and Average Pooling techniques ?

Max pooling and average pooling are techniques used in Convolutional Neural Networks (CNNs) to reduce the spatial dimensions of feature maps while retaining important information. Here's a breakdown of each:

Max Pooling

- **Purpose:** Reduces the spatial dimensions of a feature map while preserving the most important information.
- **Operation:** For each region (or window) of the feature map, max pooling selects the maximum value. This helps retain the most significant features while reducing dimensionality.
- **Example:** Consider a 2x2 max pooling operation on the following 4x4 feature map:

$$\begin{bmatrix} 1 & 3 & 2 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

Applying 2x2 max pooling with a stride of 2:

$$\begin{bmatrix} 6 & 8 \\ 14 & 16 \end{bmatrix}$$

Here, each value in the output matrix is the maximum value from a 2x2 region of the input matrix.

Average Pooling

- **Purpose**: Reduces the spatial dimensions of a feature map while averaging the values in each region.
- **Operation**: For each region (or window) of the feature map, average pooling calculates the average value. This helps in reducing the size while providing a smoother approximation of the feature map.
- **Example**: Consider the same 4x4 feature map:

$$\begin{bmatrix} 1 & 3 & 2 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

Applying 2x2 average pooling with a stride of 2:

$$\begin{bmatrix} 3.75 & 6.75 \\ 11.75 & 14.75 \end{bmatrix}$$

Here, each value in the output matrix is the average value from a 2x2 region of the input matrix.

Comparison

- **Max Pooling**:
  - **Pros**: Captures the most significant feature in each region, which can help in preserving important details.
  - **Cons**: May discard useful information that is not the maximum value in the region.
- **Average Pooling**:
  - **Pros**: Provides a more generalized representation of the region, which can smooth out the feature map.
  - **Cons**: May average out important features and details.

Both techniques are used to reduce computational complexity and overfitting by reducing the spatial dimensions of feature maps, but they do so in different ways. The choice between max pooling and average pooling often depends on the specific requirements of the task and the nature of the data.

79. Explain Flatten Layer, Padding, Stride, Filter, Kernel Size, Dropout Layer, Optimizer (Adam) ?

1. Flatten Layer

- **Purpose**: Converts a multi-dimensional tensor (e.g., the output of a convolutional layer) into a one-dimensional vector.
- **Operation**: It is typically used before passing the data to fully connected (dense) layers. For example, if the output of a convolutional layer is a 3D tensor with shape `(height, width, channels)`, the flatten layer will transform it into a 1D vector with shape `(height * width * channels)`.

2. Padding

- **Purpose**: Controls the spatial dimensions of the output feature map by adding extra pixels around the border of the input feature map.
- **Types**:
  - **Valid Padding**: No padding is added. The output size is smaller than the input size.
  - **Same Padding**: Padding is added so that the output size is the same as the input size (if stride is 1).

3. Stride

- **Purpose**: Determines how the convolutional filter moves across the input feature map.
- **Operation**: A stride of `1` means the filter moves one pixel at a time, while a stride of `2` means the filter moves two pixels at a time. Larger strides reduce the spatial dimensions of the output feature map.

4. Filter (Kernel)

- **Purpose**: The small matrix used to detect specific features in the input feature map.
- **Operation**: Each filter is applied across the entire input feature map to produce a new feature map. For example, a `3x3` filter slides over the input and performs element-wise multiplication and summation.

5. Kernel Size

- **Purpose**: Defines the dimensions of the convolutional filter.
- **Examples**: Common kernel sizes are `3x3`, `5x5`, etc. A `3x3` kernel means the filter has a width and height of `3` pixels.

6. Dropout Layer

- **Purpose**: Helps prevent overfitting by randomly setting a fraction of input units to `0` during training.
- **Operation**: During training, dropout randomly selects a subset of neurons to be dropped out, which helps the model generalize better by not relying too heavily on any particular neuron.

7. Optimizers (Adam)

- **Purpose**: Algorithms that adjust the weights of the neural network during training to minimize the loss function.
- **Adam Optimizer**:
    - **Features**: Combines the advantages of two other optimizers: AdaGrad and RMSProp. It uses adaptive learning rates for each parameter and maintains a moving average of the gradients and their squares.
    - **Mathematical Operation**:

$$\mathrm{m}_t = \beta_1 \mathrm{m}_{t-1} + (1-\beta_1)\nabla_\theta J(\theta)$$

$$\mathrm{v}_t = \beta_2 \mathrm{v}_{t-1} + (1-\beta_2)(\nabla_\theta J(\theta))^2$$

$$\hat{\mathrm{m}}_t = \frac{\mathrm{m}_t}{1-\beta_1^t}$$

$$\hat{\mathrm{v}}_t = \frac{\mathrm{v}_t}{1-\beta_2^t}$$

$$\theta = \theta - \frac{\eta \hat{\mathrm{m}}_t}{\sqrt{\hat{\mathrm{v}}_t} + \epsilon}$$

Where m_t and v_t are the first and second moment estimates, β_1 and β_2 are decay rates, η is the learning rate, and ϵ is a small constant to prevent division by zero.

80. Explain Dense Layer and Hidden Layer ?

Dense Layer (Fully Connected Layer)

- **Purpose**: Connects every neuron in the layer to every neuron in the previous layer.
- **Operation**: Each neuron in a dense layer receives input from all neurons of the previous layer, performs a weighted sum, and applies an activation function. It is typically used towards the end of a CNN to perform classification or regression.

Hidden Layer

- **Purpose**: Layers between the input layer and output layer in a neural network.
- **Operation**: Hidden layers perform transformations on the input data through weights and activation functions. In CNNs, hidden layers include convolutional layers, pooling layers, and other types of layers that process data before the final classification or regression.

81. Explain sigmoid and ReLU activation functions ? Why is ReLU commonly used ?

Definition : The sigmoid function is a type activation function defined as :

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

where eee is the base of the natural logarithm.

Explanation:

- **Range**: Outputs values between 0 and 1.
- **Shape**: The function has an S-shaped curve, also known as a sigmoid curve.
- **Use Case**: Often used in binary classification problems and in the output layer of logistic regression models to output probabilities.

Characteristics:

- **Gradient**: The derivative of the sigmoid function is:
  σ′(x)=σ(x)⋅(1−σ(x))\sigma'(x) = \sigma(x) \cdot (1 - \sigma(x))σ′(x)=σ(x)⋅(1−σ(x))
  This makes it relatively straightforward to compute gradients for backpropagation.

- **Vanishing Gradient Problem**: For very large or very small values of xxx, the gradient of the sigmoid function becomes very small, which can slow down learning and make the training of deep networks difficult.

ReLU Function (Rectified Linear Unit)

- **Definition**: The ReLU function is defined as:

$$\mathrm{ReLU}(x) = \max(0, x)$$

Explanation:

- **Range**: Outputs values between 0 and ∞\infty∞. For x≤0x \leq 0x≤0, it outputs 0, and for x>0x > 0x>0, it outputs xxx.
- **Shape**: The function is linear for positive values and flat (zero) for negative values.
- **Use Case**: Commonly used in hidden layers of deep neural networks because it introduces non-linearity and helps in learning complex patterns.

Characteristics:

- **Gradient**: The derivative of the ReLU function is:

$$\mathrm{ReLU}'(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

  This simple gradient allows for efficient computation and helps with faster training.

- **Avoiding Vanishing Gradients**: Unlike sigmoid, ReLU does not suffer from the vanishing gradient problem as severely, which helps in training deeper networks.
- **Dead Neurons**: A downside is that neurons can sometimes become inactive (i.e., always output 0) if they only receive negative inputs, which is referred to as the "dying ReLU" problem.

Why ReLU?

1. **Non-Linearity**: ReLU introduces non-linearity into the model, allowing the network to learn complex functions. Even though ReLU is not a smooth function, it still provides the necessary non-linearity.
2. **Computational Efficiency**: The ReLU function is computationally efficient because it involves simple thresholding at zero. This speeds up training compared to functions like sigmoid, which involve exponentiation and division.
3. **Sparsity**: ReLU can result in sparse activations, meaning that many neurons will be inactive (output 0) for a given input. This sparsity can make the network more efficient.
4. **Avoiding Vanishing Gradients**: ReLU helps mitigate the vanishing gradient problem by ensuring that gradients are either 0 or 1, which allows for faster convergence and better performance in deep networks.

In summary, while sigmoid is still used in certain contexts (like the output layer of binary classification), ReLU has become the default choice for hidden layers in many neural networks due to its simplicity, efficiency, and effectiveness in addressing issues like vanishing gradients.

82. Explain N-Gram ?

Definition: N-grams are continuous sequences of nnn items (words, characters, etc.) from a given text or speech. They are used in text processing and natural language processing (NLP) to capture context and relationships between items.

Examples:

- **Unigram**: Single word or character (e.g., "data").
- **Bigram**: Sequence of two items (e.g., "data leakage").
- **Trigram**: Sequence of three items (e.g., "data leakage prevention").

83. Explain Data Leakage ?

Definition: Data leakage occurs when information from outside the training dataset is used to create the model. This results in overly optimistic performance estimates because the model has access to information that it would not normally have in a real-world scenario.

Example: Using future information to train a model predicting stock prices.

84. Explain Silhouette Score (Unsupervised) ?

Definition: The silhouette score is a metric used to evaluate the quality of clustering in unsupervised learning. It measures how similar each data point is to its own cluster compared to other clusters.

Calculation: For each data point, the silhouette score is calculated as:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

where a(i)a(i)a(i) is the average distance between the point and all other points in its own cluster, and b(i)b(i)b(i) is the minimum average distance between the point and all points in the nearest cluster.

Range: The score ranges from -1 to 1, where a higher value indicates better clustering.

85. Explain Dimensionality Reduction (Feature Selection) ?

Definition: Dimensionality reduction involves reducing the number of features (dimensions) in a dataset while retaining as much information as possible.

Techniques:

- **Feature Selection**: Selecting a subset of the most relevant features (e.g., using methods like Recursive Feature Elimination (RFE)).
- **Feature Extraction**: Creating new features from the original ones (e.g., using Principal Component Analysis (PCA)).

86. What is Hyperparameter ?

Definition: Hyperparameters are parameters that are set before the learning process begins and are not learned from the data. They control the learning process and affect model performance.

Examples: Learning rate, number of hidden layers, number of trees in a random forest, etc.

87. Why Remove Stopwords ?

Definition: Stopwords are common words (e.g., "the", "is", "and") that are often removed from text data because they do not provide significant information for text analysis.

Reason: Removing stopwords can help reduce dimensionality and noise in the data, leading to more meaningful analysis and improved model performance.

88. Explain Euclidean distance and Manhattan Distance ?

Euclidean Distance

Definition: Euclidean distance is the straight-line distance between two points in a Euclidean space.

Formula: For two points (x_1,y_1) and (x_2,y_2) the Euclidean distance is:

$$\text{distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Manhattan Distance

- **Definition**: Manhattan distance, also known as L1 distance or taxicab distance, is the sum of the absolute differences of their coordinates.
- **Formula**: For two points (x_1,y_1) and (x_2,y_2) the Manhattan distance is:

$$\text{distance} = |x_2 - x_1| + |y_2 - y_1|$$

89. Explain Gini Impurity, Information Gain, Entropy ?

Gini Impurity

- **Definition**: Gini impurity is a metric used in decision trees to measure the impurity of a dataset. It is used to decide which feature to split on at each node.
- **Formula**: For a dataset with classes C1,C2,…,CkC_1, C_2, \ldots, C_kC1,C2,…,Ck, the Gini impurity is:

$$Gini = 1 - \sum_{i=1}^{k} p_i^2$$

Information Gain

- **Definition**: Information gain measures how much information a feature provides about the target variable. It is used in decision trees to select the feature that results in the highest reduction in entropy.
- **Formula**: Information gain is calculated as:

$$IG = \text{Entropy}(parent) - \text{Weighted Average Entropy}(children)$$

Entropy

- **Definition**: Entropy is a measure of the uncertainty or randomness in a dataset. It quantifies the amount of information required to describe the dataset.
- **Formula**: For a dataset with classes $C_1, C_2, \ldots, C_k$ the entropy is:

$$Entropy = -\sum_{i=1}^{k} p_i \log_2(p_i)$$

Where p_i is the probability of each class i.

90. Explain Straight Line Equation ?

Straight Line Equation

- Definition: The equation of a straight line in a 2D Cartesian coordinate system can be expressed in several forms. The most common form is the slope-intercept form:

  y=mx+b
- where:
  - y is the dependent variable (output).
  - x is the independent variable (input).
  - m is the slope of the line (rate of change).
  - b is the y-intercept (value of y when x=0).

91. Explain Logistic Regression ?

Definition: Logistic regression is a statistical method for binary classification that models the probability of a binary outcome based on one or more predictor variables.

Equation: The logistic regression model uses the logistic function (sigmoid function) to predict probabilities:

$$P(Y=1|X) = \frac{1}{1+e^{-(\beta_0+\beta_1 X)}}$$

where:

- $P(Y=1|X)$ is the probability of the dependent variable Y being 1 given X.
- β_0 is the intercept.
- β_1 is the coefficient of the predictor variable X.

The model predicts the probability that Y=1, and a threshold (usually 0.5) is used to classify Y into 0 or 1.

92. Explain Bagging and Boosting ?

Bagging (Bootstrap Aggregating):

- **Definition:** Bagging is an ensemble learning technique that combines predictions from multiple models to improve accuracy and reduce overfitting.
- **Example:** Random Forest is a popular bagging algorithm that creates multiple decision trees from different subsets of the data and averages their predictions.

Boosting:

- **Definition:** Boosting is an ensemble learning technique that sequentially trains models, with each new model focusing on correcting errors made by the previous models.
- **Example:** AdaBoost and Gradient Boosting are popular boosting algorithms that combine multiple weak learners to create a strong learner.

93. Explain SVM Kernels (Soft Margin and Hard Margin) ?

- **Support Vector Machine (SVM)**: SVMs are used for classification and regression tasks. They find the optimal hyperplane that maximizes the margin between classes.
- **Hard Margin SVM**: Assumes that the data is linearly separable and finds a hyperplane that perfectly separates the classes with no errors. It may not be suitable for noisy data.
- **Soft Margin SVM**: Allows some misclassification to handle non-linearly separable data and noisy datasets. It introduces a penalty for misclassified points to balance margin maximization and classification errors.
- **Kernels**: Functions that transform the input data into higher-dimensional space to make it easier to find a separating hyperplane.
  - **Linear Kernel**: $K(x,x')=x^T x'$
  - **Polynomial Kernel**: $K(x,x')=(x^T x'+c)^d$
  - **Radial Basis Function (RBF) Kernel**: $K(x,x')=e^{-\gamma \| x-x' \|}$

94. What is GridSearchCV ?

Definition: Grid Search with Cross-Validation (CV) is a hyperparameter tuning technique that exhaustively searches through a specified subset of hyperparameters and evaluates model performance using cross-validation.

Example: If you have hyperparameters C\text{C}C and gamma\text{gamma}gamma for an SVM model, grid search will test different combinations of these hyperparameters and use cross-validation to determine the best set.

95. What is RMSE , Absolute Error, Cost Function ?

Root Mean Squared Error (RMSE)

Definition: RMSE is a metric used to measure the average magnitude of the errors between predicted and actual values. It gives a sense of how well the model fits the data.

Formula:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

where y_i is the actual value, $y\hat{}_i$ is the predicted value, and n is the number of observations.

Absolute Error

- **Definition:** Absolute Error measures the difference between the predicted value and the actual value, without considering the direction (positive or negative).
- **Formula:**

$$\text{Absolute Error} = |y_i - \hat{y}_i|$$

Cost Function

- **Definition**: A cost function (or loss function) is a measure of how well a model's predictions match the actual values. It quantifies the error and guides the optimization process to improve the model.

- **Examples**:
  - **Mean Squared Error (MSE)**: For regression tasks.
  - **Cross-Entropy Loss**: For classification tasks.